

Macroprudential Regulation, Financial Stability, and Economic Growth*

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Abstract

How does macroprudential regulation affect economic growth and welfare? We develop an endogenous growth model in which financial crises arise from Fisherian debt deflation. A price-dependent collateral constraint and technological spillovers generate negative pecuniary externalities and physical capital externalities, respectively. We find that, for a wide range of parameter values, macroprudential regulation that corrects only these pecuniary externalities reduces economic growth. Such regulation also lowers welfare because the costs of slower growth outweigh the benefits of greater financial stability. Finally, we show that whether the decentralized economy exhibits over- or underborrowing and over- or underinvestment depends on the relative distortions in capital demand and supply induced by the two externalities.

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1 Introduction

One of the old, big questions in macro-finance is whether we should introduce financial regulation to improve social welfare. The discussion dates back to [Bagehot \(1873\)](#) and [Schumpeter \(1911\)](#), and this question has gained renewed interest since the 2008-2009 Global Financial Crisis (GFC) in the context of macroprudential regulation.¹ If there are no frictions in the economy, then the First and Second Welfare Theorems hold, and therefore there is no room for policy interventions. However, the existing literature emphasizes that financial frictions play an important role during financial crises, and these frictions provide a rationale for macroprudential regulation (e.g., [Lorenzoni \(2008\)](#), [J. Bianchi \(2011\)](#), [Jeanne and Korinek \(2019\)](#)).² Their main finding is that a harmful pecuniary externality that arises from a price-dependent financial constraint induces overborrowing by private agents, and ex ante macroprudential restrictions on borrowing improve social welfare by mitigating adverse financial amplification during crises.

An important aspect the existing literature misses is the effect on economic growth. Even though macroprudential regulation contributes to financial stability, it might significantly hurt economic growth, thereby creating the possibility that macroprudential regulation is welfare-reducing overall. More concretely, in the presence of a harmful pecuniary externality and a growth externality, macroprudential regulation that corrects pecuniary externalities might not necessarily be welfare-improving, in contrast to the finding of the existing literature.³ In fact, several policymakers and economists were concerned about the potential negative effect on economic growth when macroprudential regulation was introduced.⁴ More fundamentally, it is not obvious whether macroprudential regulation negatively affects economic growth because it could also in principle have a positive effect on growth.⁵

The aim of this paper is to clarify the relationship between economic growth and macroprudential regulation. In particular, this paper aims to provide a theoretical model suitable for answering the following two questions. First, what is the impact of macroprudential regulation on economic growth? Second, what are the welfare implications of macroprudential regulation when we take into account its impact on growth?

For the first question, we will first formalize the positive and negative impacts that macroprudential regulation might have on economic growth. On the one hand, macroprudential

¹Examples of macroprudential regulation include caps on loan-to-value (LTV) ratio, caps on debt-to-income (DTI) ratio, countercyclical capital requirement, dynamic provisioning, and so on.

²A complementary strand of literature justifies macroprudential interventions based on aggregate demand externalities in the presence of nominal rigidities; see [Farhi and Werning \(2016\)](#), [Korinek and Simsek \(2016\)](#), and [Schmitt-Grohé and Uribe \(2016\)](#).

³Following the approach taken in the literature, in this paper, macroprudential regulation is modeled as a Pigouvian tax that corrects pecuniary externalities and is used in normal times (i.e., before a financial crisis occurs).

⁴[Lo and Rogoff \(2015\)](#) summarize the hypotheses for the sluggish economic growth after the GFC, including the argument based on excessive financial regulation.

⁵The possibility of the growth-enhancing effect of macroprudential regulation is especially emphasized by the Fed and ECB. See, e.g., [Yellen \(2017\)](#) and [Constâncio \(2015\)](#).

regulation would hinder economic growth. This is because it restricts borrowing by firms, thus reducing productive investment, and thereby lowering economic growth.⁶ On the other hand, macroprudential regulation would have a beneficial impact on economic growth by reducing uncertainty in the macroeconomy. Macroprudential policies, which are designed to mitigate the likelihood and severity of financial crises, are expected to reduce macroeconomic volatility. The decline in volatility, or uncertainty, then reduces the option value of waiting in the presence of irreversibility, thereby increasing investment and growth.⁷ Having formalized the possible effects, we will then analyze whether the growth-reducing effect exceeds the growth-enhancing effect in a well-calibrated model.

For the second question, we will examine whether the welfare-reducing effect of macroprudential regulation outweighs the welfare-improving effect. On the one hand, if the growth-reducing effect exceeds the growth-enhancing effect, then macroprudential regulation would have a negative impact on social welfare by reducing economic growth. On the other hand, macroprudential regulation decreases the likelihood of a financial crisis and mitigates the severity of a crisis. This stabilization of the financial system would have a positive impact on social welfare. Therefore, there might be a cost of hindering growth in the long run whereas there is a benefit of securing financial stability in the short/medium run. Importantly, the existing literature abstracts from the welfare-reducing effect through economic growth and obtain the welfare-improving result of macroprudential regulation. Our claim is that once we take into account the effect on economic growth, it is not obvious whether macroprudential regulation always improves social welfare.

Our model has three key ingredients. First, in the absence of state-contingent claims, a collateral constraint à la [Kiyotaki and Moore \(1997\)](#) introduces harmful pecuniary externalities and triggers a financial amplification mechanism similar to the debt-deflation mechanism originally proposed in [Fisher \(1933\)](#). This financial amplification mechanism allows the model to produce the nonlinear dynamics of financial crises observed in the data (e.g., [Mendoza \(2010\)](#)). Second, we introduce technological spillovers from capital accumulation, as in [Romer \(1986\)](#), to generate endogenous growth. Introducing growth externalities into a model is important because, without it, there is only one externality in a model and macroprudential regulation correcting that externality would improve social welfare by construction. The interesting situation in which macroprudential regulation correcting the pecuniary externality could either improve or reduce social welfare arises when there is another externality in a model. Third, we introduce partial irreversibility of investment to capture the growth-enhancing effect of macroprudential

⁶For example, [Ayyagari, Beck, and Martinez Peria \(2018\)](#) provide an empirical account of a significant and negative association between borrower-targeted macroprudential policies and firms' long-term financing growth.

⁷See [Dixit and Pindyck \(1994\)](#) for a comprehensive treatment on uncertainty and investment.

regulation explained above.⁸⁹

We start by characterizing the decentralized equilibrium in which private agents take prices and aggregate variables as given. Because of the Fisherian debt-deflation mechanism, the economy experiences severe financial crises when the collateral constraint binds. Comparative statics shows that private agents increase their investment as the output volatility decreases. Next, we characterize the social planner's equilibrium in which a constrained social planner internalizes the externalities that are present in the decentralized economy. Then, we compare the decentralized and social planner's equilibria. Lastly, we evaluate the effectiveness of macroprudential regulation in terms of its impact on economic growth, financial stability, and welfare gains.

In our model, there are two externalities in the decentralized economy: pecuniary externalities and physical capital externalities (growth externalities). These two externalities create a wedge between the privately and socially optimal allocations. We show that this wedge can be expressed as a sum of two externalities and identify which part is induced by pecuniary externalities or physical capital externalities. This decomposition is particularly useful in analyzing the effect of macroprudential regulation that corrects only pecuniary externalities. In practice, macroprudential regulation aims to contain systemic risk in financial markets and is motivated by pecuniary externalities that stem from financial frictions and amplify fire sales of assets in downturns, not by growth externalities.¹⁰

Our main result is that for a large range of parameter values, macroprudential regulation that corrects only pecuniary externalities has a negative impact on growth and is welfare-reducing overall. In our model, the growth-reducing effect of macroprudential regulation exceeds the growth-enhancing effect, thereby hindering economic growth. The regulation, on the other hand, reduces the likelihood of financial crises and mitigates the severity of the crises. Therefore, there is a trade-off between economic growth and financial stability. It turns out that the cost of hindering growth outweighs the benefit of financial stability, leading to a welfare-reducing result. This result contrasts sharply with the existing literature that emphasizes welfare improvement via macroprudential regulation.

Moreover, we show that whether the decentralized economy exhibits over-/under-borrowing or over-/under-investment depends on the relative distortions in the capital demand and supply induced by the two externalities. Specifically, the interaction of the following two forces in the capital market plays a crucial role: (i) the force to reduce borrowing and thus decrease the capital demand in the planner's allocation, induced by pecuniary externalities

⁸Recall that the growth-enhancing effect is via the reduction in macroeconomic volatility. Thus, to capture this effect in a model, we need to connect the agents' decisions on productive investment with the changes in volatility (or uncertainty). One way to do this is to introduce the irreversibility of investment.

⁹Another possible way to formalize the positive impact of macroprudential regulation on economic growth is the diminishing returns to investment due to [Barlevy \(2004\)](#). We will come back to this point at the end of this paper.

¹⁰See, for instance, [FSB-IMF-BIS \(2011\)](#) or [Claessens \(2015\)](#).

of the private agent's borrowing decisions, and (ii) the force that tends to increase the capital supply, induced by physical capital externalities and pecuniary externalities of private agents' investment decisions. Hence, the decentralized economy in our model does not always exhibit over-borrowing or over-investment, in contrast to the existing literature. In particular, we find that the decentralized economy exhibits under-borrowing and under-investment relative to the social planner's equilibrium for a large range of parameter values.

Related Literature. This paper belongs to a growing body of literature on the role of macroprudential policies in mitigating financial crises, based on pecuniary externalities. This literature emphasizes over-borrowing in a decentralized economy, and that macroprudential policies that reduce borrowing are welfare-improving (e.g., [Lorenzoni \(2008\)](#), [J. Bianchi \(2011\)](#), [Schmitt-Grohé and Uribe \(2017\)](#), [J. Bianchi and Mendoza \(2018\)](#), [Dávila and Korinek \(2018\)](#), [Jeanne and Korinek \(2019\)](#)).¹¹ We contribute to this literature by examining the effect of macroprudential regulation on economic growth. We show that there is a trade-off between economic growth and financial stability and that macroprudential regulation is not necessarily welfare-improving if the cost of hindering economic growth is significant. Furthermore, we show that whether a decentralized economy exhibits over-borrowing or over-investment depends on the relative distortions in capital demand and supply.

This research is also related to a growing literature on the interactions between short-run fluctuations and long-run growth, following the lead of [Comin and Gertler \(2006\)](#).¹² A recent set of papers examines the causes and long-term impacts of the Great Recession and the financial crises (e.g., [Benigno and Fornaro \(2018\)](#), [Anzoategui, Comin, Gertler, and Martinez \(2019\)](#), [F. Bianchi, Kung, and Morales \(2019\)](#), [Guerron-Quintana and Jinnai \(2019\)](#), [Ikeda and Kurozumi \(2019\)](#), [Queralto \(2020\)](#)). In terms of macroeconomic policies, while the existing literature focuses on monetary policy, this paper focuses on macroprudential regulation. Moreover, this paper shows that macroprudential regulation which aims at stabilizing the nonlinear short-/medium-run fluctuations due to financial crises can have a negative impact on long-run economic growth by constraining the productive investment.

This research also relates to the literature on the relationship between financial development and economic growth. A number of studies have emphasized the causal link going from financial development, measured by the credit-to-GDP ratio, to economic growth (e.g., [King and Levine \(1993\)](#), [Levine and Zervos \(1998\)](#), [Rajan and Zingales \(1998\)](#), [Beck, Levine, and Loayza](#)

¹¹Notable exceptions include [Benigno, Chen, Otrok, Rebucci, and Young \(2013\)](#) and [Schmitt-Grohé and Uribe \(2018\)](#), both of whom find under-borrowing, rather than over-borrowing, in a decentralized economy. Moreover, [Benigno et al. \(2013\)](#) point out that macroprudential regulation can be welfare-reducing. However, in contrast to this paper, neither considered investment or economic growth; thus, the sources of their results are different from ours.

¹²The empirical result on the relationship between the business-cycle volatility and economic growth is mixed. While [Ramey and Ramey \(1995\)](#), [Imbs \(2007\)](#), and [Koren and Tenreyro \(2007\)](#) support a negative relationship, [Ranciére, Tornell, and Westermann \(2008\)](#) support a positive relationship.

(2000); see also Levine (2005) for surveys.) Macroprudential regulation reduces the credit in the economy and, as a result, is expected to have a negative impact on growth, according to the finding of this literature. However, as mentioned above, macroprudential regulation might also have a growth-enhancing effect by reducing volatility in the macroeconomy. In this sense, this research lies at the intersection of the two sets of literature: ‘finance and growth’ and ‘volatility and growth.’

Outline. Section 2 describes the model environment, characterizes the decentralized equilibrium, and conducts comparative statics. In Section 3, we first characterize the constrained efficient equilibrium and show the decomposition of the wedge induced by the two externalities. We then discuss how the interaction of two externalities affects capital demand and supply, leading to over-/under-borrowing or over-/under-investment in the decentralized economy. Lastly, we present a quantitative analysis and illustrate our main result on the impact of macroprudential regulation on growth and its welfare implications. Section 4 concludes.

2 Baseline Model

Our baseline model describes financial crises driven by Fisherian debt deflation in a three-period economy with collateral constraints, learning-by-doing, and asymmetric capital adjustment costs.¹³ In Section 2.1, we describe the model environment. In Section 2.2, we characterize the decentralized equilibrium. In Section 2.3, we conduct comparative statics to examine the properties of the decentralized equilibrium.

2.1 Environment

Time is discrete, and there are three periods $t \in \{0, 1, 2\}$. The economy is populated by two types of agents: a continuum of atomistic, identical entrepreneurs with measure unity, and a continuum of competitive capital goods producers with measure unity. There are two types of goods: a homogeneous consumption good, which serves as the numéraire, and a capital good. The only source of uncertainty is the output volatility in period 1. We denote by $s \in S$ the state of nature realized at the beginning of period 1, where S is the set of all possible states. For simplicity, we restrict our attention to the case where there are two states of nature, such that $S = \{h, l\}$. We further assume that state h is realized with probability p and that state l is realized with probability $1 - p$.

¹³The model that will be used in the rest of this paper is a three-period model although we recognize that an infinite-horizon model would be more suitable for answering the research questions. The main purpose of the three-period model is to understand the behavior of the model, check whether the mechanisms we expected are indeed present in the model, and build intuitions of the results. We will come back to this point at the end of this paper.

Preferences. The preference of the representative entrepreneur is given by

$$U = u(c_0) + \mathbb{E}_0 [\beta u(c_1) + \beta^2 u(c_2)] \quad (1)$$

where the period utility function $u(\cdot)$ is twice-continuously differentiable, strictly increasing, and strictly concave, $\mathbb{E}_0$ denotes a mathematical expectation conditional on period-0 information, and $\beta \in (0, 1]$ denotes a subjective discount factor.

Learning-by-Doing. We assume that the aggregate productivity level in the economy $A_t(s)$ is endogenous and takes the following form:

$$A_t(s) = A \bar{k}_t^{1-\alpha}(s)$$

where $A > 0$ is a scale parameter, and variables with bars represent aggregate variables, or “market averages,” taken as given by the representative agent but equal to the representative agent’s choices in equilibrium. Following [Arrow \(1962\)](#), [Sheshinski \(1967\)](#), and [Romer \(1986\)](#), we interpret the assumption that the aggregate productivity level is proportional to the aggregate capital stock as “learning-by-doing”; that is, technological progress is a by-product of capital accumulation, and anyone can enjoy this technological advancement without preventing others from using the same knowledge. Importantly, the details behind the endogenous productivity mechanisms are largely irrelevant for our purposes, and we could alternatively assume the human capital investment of [Lucas \(1988\)](#), the expanding variety framework of [Romer \(1990\)](#), and the Schumpeterian growth framework of [Aghion and Howitt \(1992\)](#).

Technology. In periods 1 and 2, the representative entrepreneur employs capital to produce consumption goods using the following production technology:

$$\begin{aligned} y_1(s) &= Z(s) A_1 k_1^\alpha \\ y_2(s) &= A k_2(s) \end{aligned}$$

where the representative agent takes $A_t(s)$ as given, and $Z(s)$ is a random variable representing the output volatility, which depends on the state of nature $s \in S$. For simplicity, we assume that the aggregate productivity level in period 2 is determined exogenously and is constant, while that of period 1 is endogenous.¹⁴¹⁵ Therefore, in this three-period model where technological progress is a by-product of capital accumulation, productivity growth comes only from period-0 investment and thus the variable of our main interest is period-0 investment.¹⁶

¹⁴This assumption is not crucial for our results. The qualitative implications are preserved when we alternatively assume that the aggregate productivity level in period 2 is also endogenous. However, eliminating physical capital externalities in period 2 makes the normative implications in Section 3 much more intuitive.

¹⁵Note that the production technologies in the two periods are essentially the same and linear in capital, given that the aggregate variables are equal to the representative agent’s choices in equilibrium.

¹⁶Economic growth in this three-period model means the change in aggregate productivity level in period 1, A_1 .

Output Volatility. The random variable $Z(s)$ is assumed to take two values, such that

$$Z(s) = \begin{cases} Z^h & \text{if } s = h \\ Z^l & \text{if } s = l \end{cases}$$

where $Z^h > 1 > Z^l$. We introduce this stochastic term for the following reason: As shown in the literature, macroprudential regulation correcting the credit externality reduces the long-run probability of a financial crisis and ameliorates the severity of a crisis. As a result, macroprudential regulation is expected to reduce the volatility of output. It is this output volatility that the above stochastic term is meant to capture in this simple three-period model. Thus, in this three-period model, we are mainly interested in how the changes in output volatility affect the period-0 investment decision, which in turn affects productivity growth.

Budget Constraints. In periods 0 and 1, the representative entrepreneur can issue a one-period, non-state-contingent debt at the fixed gross interest rate R , determined exogenously in the world market. In period 1, his ability to roll over debt may be affected by a collateral constraint. There is also a market to trade capital with capital goods producers at prices q_0 and $q_1(s)$ in periods 0 and 1, respectively. The purchase of new capital could be interpreted as making investment, which contributes to the subsequent period's production. Because we assume that capital goods producers live for the first two periods, there is no capital market in period 2, and we assume that capital fully depreciates at the end of period 2.

Under these assumptions, the budget constraints of the representative entrepreneur are given by

$$c_0 + q_0 k_1 = \frac{d_1}{R} + q_0(1 - \delta)k_0 + \Pi_0 \quad (2)$$

$$c_1(s) + q_1(s)k_2(s) + d_1 = y_1(s) + \frac{d_2(s)}{R} + q_1(s)(1 - \delta)k_1 + \Pi_1(s), \quad \forall s \quad (3)$$

$$c_2(s) + d_2(s) = y_2(s), \quad \forall s \quad (4)$$

where $d_{t+1}(s)$ is the debt to be repaid in period $t + 1$, $\Pi_t(s)$ represents the profits of capital goods producers in period t , and $\delta \in [0, 1]$ is the depreciation rate.

Collateral Constraints. In period 1, the entrepreneur faces a collateral constraint that limits debt, not to exceed a fraction κ (< 1) of the market value of the beginning-of-period capital holdings:

$$\frac{d_2(s)}{R} \leq \kappa q_1(s)k_1, \quad \forall s \quad (5)$$

The micro-foundation for this constraint (5) is that the entrepreneur could walk away from his debt, following which creditors can seize a fraction κ of his collateral and sell it in the capital market. The form of this constraint follows [Jeanne and Korinek \(2010\)](#) and [J. Bianchi and Mendoza \(2018\)](#) and is common in the literature on macroprudential regulation and financial

crises.¹⁷

Capital Goods Producers

We now turn to the setup of capital goods producers. Competitive capital goods producers live for the first two periods $t \in \{0,1\}$ and are owned by the entrepreneur. The main role of capital goods producers is to create new capital in the economy. When creating new capital, capital goods producers are assumed to incur the capital adjustment cost $\Psi(n_t(s)/k_t)k_t$, where $n_t(s) := k_{t+1}(s) - k_t$ denotes net investment.¹⁸

Profits. Capital goods producers purchase undepreciated capital $(1 - \delta)k_t$ from the entrepreneur and sell him new capital $k_{t+1}(s)$ at a price $q_t(s)$. New capital is produced by making investment $i_t(s)$. Then the profits of the capital goods producers are given by

$$\Pi_t(s) = q_t(s)k_{t+1}(s) - q_t(s)(1 - \delta)k_t - i_t(s) \quad (6)$$

where $i_t(s) := \delta k_t + n_t(s) + \Psi(n_t(s)/k_t)k_t$ denotes gross investment.

Partial Irreversibility. The specification of capital adjustment costs is a key element of our analysis. Modifying the asymmetric wage adjustment costs developed by [Fahr and Smets \(2010\)](#), we introduce the following *asymmetric* capital adjustment cost function:

$$\Psi\left(\frac{k_{t+1}(s) - k_t}{k_t}\right) = \frac{\phi - 1}{2} \left(\frac{k_{t+1}(s) - k_t}{k_t}\right)^2 + \frac{1}{\eta^2} \left(\exp\left\{-\eta \left(\frac{k_{t+1}(s) - k_t}{k_t}\right)\right\} + \eta \left(\frac{k_{t+1}(s) - k_t}{k_t}\right) - 1\right) \quad (7)$$

where the parameter $\phi > 1$ determines the degree of convexity, and $\eta > 0$ the degree of asymmetry of the capital adjustment cost. This functional form nests the standard symmetric, quadratic capital adjustment cost: if the asymmetric parameter η approaches zero ($\eta \rightarrow 0$), then

$$\lim_{\eta \rightarrow 0} \Psi\left(\frac{k_{t+1}(s) - k_t}{k_t}\right) = \frac{\phi}{2} \left(\frac{k_{t+1}(s) - k_t}{k_t}\right)^2.$$

Figure 1 illustrates the properties of the capital adjustment cost function (7). If η takes a strictly positive value, the capital adjustment cost is asymmetric and negative investment is more costly than positive investment. An increase in η raises the cost of negative investment but leaves the cost of positive investment almost unchanged.

2.2 Decentralized Equilibrium

The definition of competitive equilibrium is as follows:

¹⁷J. [Bianchi and Mendoza \(2018\)](#) show that this form of collateral constraint can be derived as an incentive compatibility constraint resulting from a limited enforcement problem.

¹⁸Following [Mendoza \(2010\)](#), the capital adjustment cost is specified in terms of net investment, rather than gross investment.

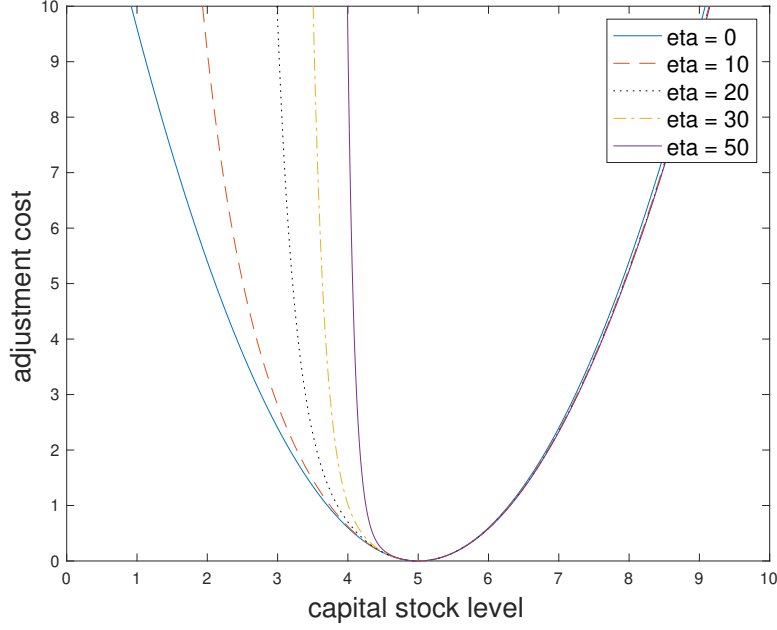


Figure 1: Asymmetric Capital Adjustment Cost Function

Notes: The convexity parameter ϕ is set as $\phi = 30$, and the state variable k_t is set as $k_t = 5$. We draw the capital adjustment cost for each asymmetry parameter η . If the capital stock level in the next period, k_{t+1} , is less than $k_t = 5$, then it corresponds to negative investment, and vice versa.

Definition 1. Given $(k_0, \{Z(s)\}_{s \in S})$, a competitive equilibrium for this economy consists of a set of period-0 choices (c_0, k_1, d_1, i_0) , period-1 choices $(\{c_1(s), c_2(s), k_2(s), d_2(s), i_1(s)\}_{s \in S})$, and prices $(q_0, \{q_1(s)\}_{s \in S})$ such that

1. The representative entrepreneur maximizes his utility (1) subject to the budget constraints (2), (3), (4) and the collateral constraint (5), taking as given prices and aggregate variables;
2. Capital goods producers maximize profits (6), taking prices as given;
3. The capital markets clear; and
4. The representative entrepreneur's choices satisfy $k_t(s) = \bar{k}_t(s), c_t(s) = \bar{c}_t(s), d_t(s) = \bar{d}_t(s)$, and $i_t(s) = \bar{i}_t(s)$, for all $s \in S$.

To simplify the analysis, we make the following parametric assumptions: $\beta = 1, \delta = 0, R = 1$. We also assume that $A > 1$. The latter assumption ensures that there is a positive investment whenever the collateral constraint does not bind.

We solve for the decentralized equilibrium via backward induction.

Period 1–2 Equilibrium. We first focus on the capital goods producers' decision problem. In

period 1, the profit maximization problem of capital goods producers is given by

$$\max_{k_2(s)} \Pi_1(s) = q_1(s)(k_2(s) - k_1) - \left[k_2(s) - k_1 + \Psi \left(\frac{k_2(s) - k_1}{k_1} \right) k_1 \right].$$

The resulting optimal investment condition is:

$$q_1(s) = 1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) \quad (8)$$

This condition states that the marginal benefit of capital investment, $q_1(s)$, is equated to the marginal cost of capital investment, $1 + \Psi'((k_2(s) - k_1)/k_1)$, at the optimum.

Next, we turn to the entrepreneur's decision problem. Following [Dávila and Korinek \(2018\)](#), we fully describe the state of the economy in period 1 by two sets of state variables: (i) the net worth in terms of consumption goods (not including the capital stock), defined by

$$m_1(s) := Z(s)A_1k_1^\alpha - d_1,$$

and (ii) the capital stock k_1 in period 1. Note that the agent's net worth fully captures the effect of uncertainty on the economy. Furthermore, $Z(s)$ is assumed to be realized at the beginning of period 1, so that there is no uncertainty when the entrepreneur makes a decision in period 1. Given the state variables in period 1, the entrepreneur's optimization problem in period 1 is given by

$$V^{de} \left(m_1(s), k_1; \bar{m}_1(s), \bar{k}_1 \right) = \left[\begin{array}{ll} \max_{c_1(s), c_2(s), k_2(s), d_2(s)} & u(c_1(s)) + u(c_2(s)) \\ \text{s.t.} & c_1(s) + q_1(s)k_2(s) = m_1(s) + d_2(s) + q_1(s)k_1 + \Pi_1(s), \\ & c_2(s) + d_2(s) = Ak_2(s), \\ & d_2(s) \leq \kappa q_1(s)k_1, \end{array} \right]$$

where we let $\lambda_1(s), \lambda_2(s), \mu(s)$ be Lagrange multipliers attached to the budget constraints in periods 1 and 2 and the collateral constraint, respectively. Note that the value function $V^{de}(\cdot)$ distinguishes between individual state variables $(m_1(s), k_1)$ and aggregate state variables $(\bar{m}_1(s), \bar{k}_1)$. This distinction matters because the representative agent takes aggregate state variables as given, but internalizes that he can affect his own state variables through his period-0 actions.

The first-order conditions and the complementary slackness conditions are:

$$u'(c_1(s)) = \lambda_1(s) \quad (9)$$

$$u'(c_2(s)) = \lambda_2(s) \quad (10)$$

$$\lambda_1(s) = \lambda_2(s) + \mu(s)$$

$$q_1(s)\lambda_1(s) = \lambda_2(s)A$$

$$\mu(s) \geq 0, \mu(s)(\kappa q_1(s)k_1 - d_2(s)) = 0$$

Combining the above conditions, we obtain the following Euler equation and asset pricing equation:

$$u'(c_1(s)) = u'(c_2(s)) + \mu(s) \quad (11)$$

$$q_1(s) = \frac{u'(c_2(s))}{u'(c_1(s))} A \quad (12)$$

The Euler equation (11) implies that when the collateral constraint is binding (i.e., $\mu(s) > 0$), the marginal utility of current consumption exceeds the marginal utility cost of reducing future consumption by an amount equal to the marginal benefit of relaxing the constraint. The asset pricing equation (12) states that the marginal cost of an extra unit of capital, $q_1(s)$, is equated to its discounted marginal benefit, that is, the marginal product of capital, A , discounted at the marginal rate of substitution, $u'(c_2(s))/u'(c_1(s))$.

Depending on the realization of the state of nature, the period 1–2 equilibrium can be divided into two cases: an *unconstrained equilibrium*, and a *constrained equilibrium*. Intuitively, if the net worth in period 1, $m_1(s)$, is sufficiently high, then the equilibrium has the same feature as in the model without the collateral constraint. However, if $m_1(s)$ is sufficiently low, then the entrepreneur faces the binding collateral constraint that triggers a financial amplification mechanism similar to the debt-deflation mechanism.

Unconstrained Period 1–2 Equilibrium

In this case, the collateral constraint does not bind and, hence, the complementary slackness condition implies that $\mu(s) = 0$. This then implies that $u'(c_1(s)) = u'(c_2(s))$, or $c_1(s) = c_2(s)$. Thus, the entrepreneur succeeds in smoothing consumption perfectly. This implies that the asset pricing equation (12) becomes

$$q_1(s) = A. \quad (13)$$

Therefore in the unconstrained period 1–2 equilibrium, the equilibrium capital price is equal to the scale parameter of aggregate productivity, A . Given that the optimal investment condition of capital goods producers is given by equation (8) and that $A > 1$ and $\Psi'(\cdot) > 0$, this implies positive investment in the unconstrained period 1–2 equilibrium.

In summary, the entrepreneur is able to smooth consumption perfectly, investment is positive, and the price of capital is stable in the unconstrained equilibrium.

Constrained Period 1–2 Equilibrium

In the constrained period 1–2 equilibrium, the collateral constraint is binding, so that $d_2(s) = \kappa q_1(s)k_1$ and $\mu(s) > 0$. This implies that $u'(c_1(s)) > u'(c_2(s))$, or $c_1(s) < c_2(s)$. Hence, the entrepreneur fails to smooth consumption, which implies that the asset pricing equation (12) now becomes

$$q_1(s) = \underbrace{\frac{u'(c_2(s))}{u'(c_1(s))}}_{<1} A. \quad (14)$$

Because the marginal rate of substitution is less than one in this case, the equilibrium capital price in the period 1–2 constrained equilibrium (14) is lower than that of the period 1–2 unconstrained equilibrium (13).

The key mechanism at work in the constrained equilibrium is the Fisherian debt-deflation mechanism, which can be intuitively described as follows. When the collateral constraint is binding, the entrepreneur responds by reducing his spending and fire-selling capital to meet the constraint (i.e., by decreasing his demand for capital). However, when he does so, he meets with capital goods producers that feature an upward-sloping capital supply curve because of the capital adjustment costs. Thus, these capital goods producers find it optimal to reduce investment, given the reduced demand for capital and the higher discount rate for the marginal product of capital in the next period. As a result, the equilibrium capital price decreases. As the value of the collateral is tied to those capital prices, such a decline further tightens the collateral constraint. This forces the entrepreneur to cut back further on spending and engage in further fire sales of capital, triggering a vicious cycle of declining borrowing capacity, falling spending, and collapsing capital prices.

In summary, in the constrained period 1–2 equilibrium, the economy experiences a financial crisis in which the severity is amplified by Fisherian debt deflation, that is, a downward spiral of declining consumption, dropping investment, and collapsing capital prices.

Having described the properties of the unconstrained / constrained period 1–2 equilibria, we now turn to the following question: *Under what conditions does the economy fall into the constrained period 1–2 equilibrium?* A collateral constraint does not bind if and only if

$$\underbrace{m_1(s)}_{=Z(s)A_1k_1^\alpha - d_1} > Ak_2^u + (k_2^u - k_1) + \Psi\left(\frac{k_2^u - k_1}{k_1}\right)k_1 - 2\kappa Ak_1. \quad (15)$$

where the superscript u denotes allocations in the unconstrained period 1–2 equilibrium. Intuitively, the economy experiences a financial crisis in period 1 if the realization of the entrepreneur's net worth $m_1(s)$ is sufficiently low. Note that this threshold condition depends on the period-0 choices d_1 and k_1 , and on the output volatility $Z(s)$. We can easily verify that high debt d_1 and the realization of low values of $Z(s)$ make the economy likely to experience the crisis. However, it is ambiguous how the investment decisions in period 0, k_1 , affect the threshold condition. This is because it is, in general, difficult to determine whether the right-hand side of equation (15) decreases when k_1 increases, given the presence of asymmetric capital adjustment costs.

Period 0 Equilibrium. As in the period 1–2 equilibrium, the profit maximization problem of capital goods producers in period 0 is given by

$$\max_{k_1} \Pi_0 = q_0(k_1 - k_0) - \left[k_1 - k_0 + \Psi\left(\frac{k_1 - k_0}{k_1}\right)k_1 \right].$$

The resulting optimal investment condition is:

$$q_0 = 1 + \Psi' \left(\frac{k_1 - k_0}{k_0} \right) \quad (16)$$

Next, we describe the decision problem of the representative entrepreneur in period 0. The entrepreneur's utility maximization problem in period 0 is given by

$$\begin{aligned} \max_{c_0, k_1, d_1} \quad & u(c_0) + \mathbb{E}_0 \left[V^{de} \left(Z A_1 k_1^\alpha - d_1, k_1; \bar{m}_1, \bar{k}_1 \right) \right] \\ \text{s.t.} \quad & c_0 + q_0 k_1 = d_1 + q_0 k_0 + \Pi_0. \end{aligned}$$

The first-order conditions are:

$$\begin{aligned} u'(c_0) &= \mathbb{E}_0 \left[V_m^{de} \left(m_1, k_1; \bar{m}_1, \bar{k}_1 \right) \right] \\ q_0 u'(c_0) &= \mathbb{E}_0 \left[V_m^{de} \left(m_1, k_1; \bar{m}_1, \bar{k}_1 \right) \alpha Z A_1 k_1^{\alpha-1} + V_k^{de} \left(m_1, k_1; \bar{m}_1, \bar{k}_1 \right) \right] \end{aligned}$$

where $V_m^{de}(\cdot) := \frac{\partial V^{de}(\cdot)}{\partial m_1}$ and $V_k^{de}(\cdot) := \frac{\partial V^{de}(\cdot)}{\partial k_1}$. The envelope theorem implies that $V_m^{de}(\cdot) = \lambda_1(s)$ and $V_k^{de}(\cdot) = \lambda_1(s)q_1(s) + \mu(s)\kappa q_1(s)$. Thus, using the envelope conditions, we obtain a set of Euler equation and asset pricing equation:

$$u'(c_0) = \mathbb{E}_0 [u'(c_1)] \quad (17)$$

$$q_0 u'(c_0) = \mathbb{E}_0 \left[u'(c_1) \left\{ \alpha Z A_1 k_1^{\alpha-1} + q_1 \right\} + \mu \kappa q_1 \right] \quad (18)$$

The Euler equation (17) states that the marginal benefit of borrowing in period 0, $u'(c_0)$, is equated to the expected marginal cost associated with it, $\mathbb{E}_0 [u'(c_1)]$. The asset pricing equation (18) states that the marginal cost of an extra unit of capital is equal to its discounted expected marginal benefit, which consists of the marginal product of capital $\alpha Z(s) A_1 k_1^{\alpha-1}$, the continuation value $q_1(s)$ of capital, and the benefit of relaxing the collateral constraint $\mu(s)\kappa q_1(s)$.

2.3 Comparative Statics

This section conducts comparative statics to examine the properties of the decentralized equilibrium. We illustrate how the changes in output volatility, the degree of irreversibility, and attitude toward risk affect aggregate productivity and the severity of the crises.

To this end, we assume for simplicity that Z^l is low enough that the collateral constraint binds in period 1, and that Z^h is high enough that the constraint does not bind. Then equation (17) becomes

$$\begin{aligned} u'(c_0) &= p \cdot u'(c_1(h)) + (1-p) \cdot u'(c_1(l)) \\ &= p \cdot u'(c_1^u) + (1-p) \cdot u'(c_1^c), \end{aligned} \quad (19)$$

and equation (18) becomes

$$\begin{aligned} q_0 u'(c_0) &= p \cdot \left[u'(c_1(h)) \left\{ \alpha Z(h) A_1 k_1^{\alpha-1} + q_1(h) \right\} \right] + (1-p) \cdot \left[u'(c_1(l)) \left\{ \alpha Z(l) A_1 k_1^{\alpha-1} + q_1(l) \right\} + \mu(l)\kappa q_1(l) \right] \\ &= p \cdot \left[u'(c_1^u) \left\{ \alpha Z^h A_1 k_1^{\alpha-1} + q_1^u \right\} \right] + (1-p) \cdot \left[u'(c_1^c) \left\{ \alpha Z^l A_1 k_1^{\alpha-1} + q_1^c \right\} + \mu^c \kappa q_1^c \right]. \end{aligned} \quad (20)$$

Table 1: Parameter Values for Numerical Illustration

Parameter	Value	Description
α	0.33	capital share
A	1.5	scale parameter
κ	0.4	tightness of collateral constraint
σ	2	relative risk aversion
ϕ	5	degree of convexity
η	30	degree of asymmetry

where the superscript u denotes allocations in the unconstrained period 1–2 equilibrium, and the superscript c denotes allocations in the constrained period 1–2 equilibrium.

We now present the comparative statics results by conducting simple numerical simulations. The functional form for preferences is given by

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 1.$$

Table 1 summarizes the parameter values for the baseline case. We set the capital share parameter to the standard value $\alpha = 0.33$. The CRRA coefficient is set to the standard value $\sigma = 2$. We set the scale parameter of the technology level to $A = 1.5$, set the tightness of the collateral constraint to $\kappa = 0.4$, set the degree of convexity of the capital adjustment cost function to $\phi = 5$, and set the degree of asymmetry to $\eta = 30$. In addition, we set the values of the random variable $Z(s)$ to $Z^h = 105/100$ and $Z^l = 70/100$, set the probability that the state is h to $p = 9/10$, and set the initial capital stock level to $k_0 = 5$.¹⁹

Focusing on the constrained equilibrium, we examine how the key variables, especially net investment in period 0, vary in response to the changes in output volatility, the irreversibility parameter, and relative risk aversion. Note that with the form of aggregate productivity considered in this paper, it is net investment in period 0 that affects productivity growth.

Table 2 illustrates how the changes in the volatility of $Z(s)$ influence the equilibrium variables. Here, we consider a mean-preserving increase in the volatility of $Z(s)$, in the sense that the expected value of $Z(s)$ equals 1.015 in all cases. Note that net investment in period 0 decreases as the output volatility in period 1 increases. Intuitively, private agents understand that when the collateral constraint binds in period 1, it is difficult to make negative investment because of partial irreversibility, even if the entrepreneur would like to do so for consumption smoothing. Given this, agents reduce net investment in period 0, so that they do not have to make large negative investment when the constraint binds. As the output volatility increases, the amount of negative investment needed to smooth consumption increases, because of the

¹⁹These parameters and variables are set so as to roughly match the Mexican data presented in [Mendoza \(2010\)](#); in particular, we match the following two average ratios: (i) consumption to output ratio, given by 0.65, and (ii) investment to output ratio, given by 0.172.

lower value of Z^l . Thus, net investment in period 0 decreases as the volatility increases. This negative relationship between investment and volatility provides important implications for macroprudential policies. That is, macroprudential regulation that is expected to reduce output volatility in the long run would have positive effects on investment, thereby enhancing economic growth.

Table 3 shows how the changes in the degree of asymmetry in the capital adjustment cost affect the equilibrium variables. The important point is that net investment in period 0 increases as the degree of asymmetry decreases. Intuitively, at the low value of η , even if the low value of $Z(s)$ is realized in period 1 (i.e., low net worth $m_1(s)$), the entrepreneur can make negative investment without large adjustment costs in order to smooth consumption. As a result, agents do not have to reduce investment much in period 0, in fear of inability to make large negative investment in period 1 because of partial irreversibility.

Table 4 illustrates how the changes in relative risk aversion affect the equilibrium variables. We can see that net investment in period 0 decreases as the relative risk aversion increases. As the agent is getting risk averse, then net investment in period 0 declines in fear of the realization of the low value of Z in period 1.

Table 2: Comparative Statics: Output Volatility $Z(s)$

	$Z \in \{\frac{187}{180}, \frac{80}{100}\}$	$Z \in \{\frac{47}{45}, \frac{75}{100}\}$	$Z \in \{\frac{105}{100}, \frac{70}{100}\}$	$Z \in \{\frac{19}{18}, \frac{65}{100}\}$	$Z \in \{\frac{191}{180}, \frac{60}{100}\}$
$n_0 (= k_1 - k_0)$	0.403	0.274	0.148	0.030	-0.076
$n_1^c (= k_2^c - k_1)$	-0.499	-0.492	-0.485	-0.477	-0.470
c_0	4.522	4.295	4.060	3.816	3.567
c_1^c	2.107	1.890	1.699	1.528	1.373
c_2^c	7.069	6.937	6.802	6.670	6.551
q_0	1.353	1.246	1.138	1.029	0.920
q_1^c	0.133	0.111	0.094	0.079	0.066
c/y	0.667	0.682	0.697	0.713	0.729
i/y	0.080	0.079	0.078	0.077	0.076

Notes: We consider a mean-preserving increase in volatility of $Z(s)$. Note that the probability that state h is realized is set to 9/10. The volatility of $Z(s)$ rises as we move to the right. The consumption-to-output ratio is the expected value in periods 1 and 2. The investment-to-output ratio is the expected value in period 1.

Table 3: Comparative Statics: Irreversibility Parameter η

	$\eta \rightarrow 0$	$\eta = 10$	$\eta = 20$	$\eta = 30$	$\eta = 50$	$\eta = 70$	$\eta = 100$
$n_0(=k_1 - k_0)$	0.215	0.190	0.166	0.148	0.125	0.112	0.099
$n_1^c(=k_2^c - k_1)$	-0.907	-0.740	-0.590	-0.485	-0.358	-0.285	-0.221
c_0	4.204	4.148	4.097	4.060	4.011	3.981	3.952
c_1^c	1.814	1.768	1.728	1.699	1.663	1.641	1.621
c_2^c	6.194	6.440	6.654	6.802	6.977	7.075	7.161
q_0	1.215	1.184	1.157	1.138	1.115	1.101	1.088
q_1^c	0.129	0.113	0.101	0.094	0.085	0.081	0.077
c/y	0.692	0.694	0.696	0.697	0.700	0.701	0.703
i/y	0.062	0.070	0.075	0.078	0.082	0.083	0.085

Notes: We vary the irreversibility parameter η . The degree of irreversibility increases as we move to the right. The consumption-to-output ratio is the expected value in periods 1 and 2. The investment-to-output ratio is the expected value in period 1.

Table 4: Comparative Statics: Relative Risk Aversion σ

	$\sigma \rightarrow 1$	$\sigma = 1.5$	$\sigma = 2$	$\sigma = 2.5$	$\sigma = 3$
$n_0(=k_1 - k_0)$	0.725	0.409	0.148	-0.046	-0.176
$n_1^c(=k_2^c - k_1)$	-0.475	-0.492	-0.485	-0.475	-0.467
c_0	4.623	4.301	4.060	3.865	3.706
c_1^c	1.465	1.569	1.699	1.805	1.890
c_2^c	7.186	7.033	6.802	6.602	6.464
q_0	1.613	1.358	1.138	0.952	0.797
q_1^c	0.301	0.158	0.094	0.059	0.037
<i>Average Ratios</i>					
c/y	0.648	0.667	0.697	0.713	0.724
i/y	0.078	0.078	0.078	0.078	0.078

Notes: We vary the CRRA coefficient σ . The degree of relative risk aversion increases as we move to the right. The consumption to output ratio is the expected value in periods 1 and 2. The investment to output ratio is the expected value in period 1.

3 Efficiency Analysis

The previous section analyzed the decentralized equilibrium in which private agents take aggregate variables and prices as given. This atomistic behavior of private agents generates the

following two externalities in this model. The first is pecuniary externalities. As shown in the literature (e.g., [Caballero and Krishnamurthy \(2003\)](#), [Lorenzoni \(2008\)](#), [Dávila and Korinek \(2018\)](#)), in the presence of a credit constraint à la [Kiyotaki and Moore \(1997\)](#), private agents fail to internalize the effects of their borrowing and investment decisions on the price of capital. The second is physical capital externalities (growth externalities). As in [Romer \(1986\)](#), technological spillovers from capital accumulation are assumed to be external to private agents; thus, their investment decisions involve externalities.

Following the approach taken in the literature, we now focus on the *constrained* social planner’s problem.²⁰ We formulate the social planner’s problem in a manner similar to the “primal approach” to optimal policy analysis, with the planner choosing allocations subject to resource constraints, implementability constraints, and a collateral constraint. In particular, the social planner chooses debt on behalf of the entrepreneur and investment on behalf of capital goods producers, but is assumed to lack the ability to commit to future policies. Because the asset pricing equations of the entrepreneur enter in the social planner’s problem as implementability constraints, the price of capital remains market-determined.²¹ In this setup, unlike private agents, the social planner internalizes pecuniary externalities and physical capital externalities that are present in the decentralized economy. Note that the social planner considered here is constrained in three aspects. First, the planner faces the same collateral constraint as private agents. Second, the planner takes as given the fact that the price of capital is determined competitively in the capital market. In other words, the planner does not set the price of capital, but internalizes how his borrowing and investment decisions affect it. Third, the planner lacks the ability to commit to future policies.²²

In Section 3.1, we characterize the constrained efficient equilibrium and examine the difference between the decentralized equilibrium and the constrained efficient equilibrium. We also show that the wedge induced by physical capital externalities and pecuniary externalities can be decomposed into each part. In Section 3.2, we demonstrate that the constrained efficient equilibrium can be decentralized with Pigouvian taxes on debt and capital investment. We also graphically illustrate how the planner deals with externalities and how the interaction of externalities leads to over-/under-borrowing or over-/under-investment in the decentralized economy. In Section 3.3, we provide the model’s quantitative implications by conducting simple numerical simulations and discuss our main result.

²⁰In setting up the social planner’s problem, we follow the constrained-efficiency definition of [Kehoe and Levine \(1993\)](#).

²¹Although the fact that prices enter in a *planner’s* problem seems awkward to some of the readers, this naming convention is standard in the literature (see, e.g., [Lorenzoni \(2008\)](#), [Jeanne and Korinek \(2010\)](#)).

²²See [J. Bianchi and Mendoza \(2018\)](#) for details on the time inconsistency issue of macroprudential regulation when the collateral constraint takes the form considered in this paper.

3.1 Social Planner's Equilibrium

As in the decentralized equilibrium, we solve for the social planner's equilibrium via backward induction. The social planner maximizes the utility of the entrepreneur subject to resource constraints, implementability constraints, and a collateral constraint. In particular, the social planner chooses d_{t+1} on behalf of the entrepreneur and k_{t+1} on behalf of capital goods producers. The asset pricing equations of the entrepreneur (12) and (18) enter in the planner's problem as implementability constraints.

Period 1–2 SP's Problem. In period 1, the social planner solves

$$V^{sp}(\bar{m}_1(s), \bar{k}_1) = \begin{cases} \max_{\bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), q_1(s)} & u(\bar{c}_1(s)) + u(\bar{c}_2(s)) \\ \text{s.t.} & \bar{c}_1 + \bar{k}_2(s) + \Psi\left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1}\right) \bar{k}_1 = \bar{m}_1(s) + \bar{k}_1 + \bar{d}_2(s), \\ & \bar{c}_2(s) + \bar{d}_2(s) = A\bar{k}_2(s), \\ & \bar{d}_2(s) \leq \kappa q_1(s) \bar{k}_1, \\ & q_1(s) = \frac{u'(\bar{c}_2(s))}{u'(\bar{c}_1(s))} A, \end{cases} \quad (21)$$

where $\bar{m}_1(s) = Z(s)A\bar{k}_1 - \bar{d}_1$, and we let $\lambda_1^{sp}(s), \lambda_2^{sp}(s), \mu^{sp}(s)$ be Lagrange multipliers of the social planner attached to the resource constraints in periods 1 and 2 and the collateral constraint, respectively. Note that the choice variables of the planner are aggregate variables, rather than individual variables. Thus, the planner internalizes the general equilibrium effects that are uninternalized by private agents. The production function of the social planner is given by $y_1(s) = Z(s)A\bar{k}_1$, rather than that employed by the private agent, $y_1(s) = Z(s)A_1k_1^\alpha$. This creates the difference between the marginal product of capital of the planner and that of the private agent. The first-order conditions and the complementary slackness conditions are:

$$u'(\bar{c}_1(s)) - u''(\bar{c}_1(s))q_1(s)\frac{\mu^{sp}(s)\kappa\bar{k}_1}{u'(\bar{c}_1(s))} = \lambda_1^{sp}(s) \quad (22)$$

$$u'(\bar{c}_2(s)) + u''(\bar{c}_2(s))q_1(s)\frac{\mu^{sp}(s)\kappa\bar{k}_1}{u'(\bar{c}_2(s))} = \lambda_2^{sp}(s) \quad (23)$$

$$\lambda_1^{sp}(s) = \lambda_2^{sp}(s) + \mu^{sp}(s) \quad (24)$$

$$\left[1 + \Psi'\left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1}\right)\right] \lambda_1^{sp}(s) = \lambda_2^{sp}(s)A \quad (25)$$

$$\mu^{sp}(s) \geq 0, \mu^{sp}(s)(\kappa q_1(s)\bar{k}_1 - \bar{d}_2(s)) = 0$$

The first key difference between the social planner's equilibrium and the decentralized equilibrium follows from examining equations (22) and (9) and equations (23) and (10). Equation (9) in the decentralized equilibrium states that the shadow value of wealth for the private agent is equal to the marginal utility of current consumption. On the other hand, equation (22) in the social planner's equilibrium indicates that in addition to the marginal utility of current consumption, the shadow value of wealth for the social planner adds the term that captures an additional

benefit in increasing an extra unit of current consumption, given by $-u''(\bar{c}_1(s))q_1(s)\frac{\mu^{sp}(s)\kappa\bar{k}_1}{u'(\bar{c}_1(s))}$. Note that this term is non-negative because $u''(\cdot) < 0$ by assumption and $\mu^{sp}(s) \geq 0$ by the complementary slackness condition. Moreover, the complementary slackness condition implies that this additional term takes positive values only when the collateral constraint binds in period 1, (i.e., $\mu^{sp}(s) > 0$). Therefore, this term considers how an extra unit of current consumption raises the equilibrium capital price, thereby relaxing the collateral constraint when the constraint binds already. Intuitively, the social planner internalizes that when the collateral constraint binds already in period 1, the period-1 price of capital can be pushed up to increase borrowing capacity by changing current borrowing decisions. Note that equations (22) and (9) coincide when the collateral constraint is not binding.

Equation (10) in the decentralized equilibrium and equation (23) in the social planner's equilibrium also differ. Equation (23) states that the shadow value of wealth for the planner in period 2 includes, in addition to the marginal utility of consumption in period 2, the term that captures an additional benefit in decreasing consumption in period 2, which is given by $u''(\bar{c}_2(s))q_1(s)\frac{\mu^{sp}(s)\kappa\bar{k}_1}{u'(\bar{c}_2(s))}$ and is non-positive. Note that this term is negative when the collateral constraint is binding. Intuitively, when the collateral constraint binds in period 1, the planner promises lower future consumption so as to push up period-1 capital prices and borrowing capacity. Notably, this effect is present only in this simple three-period model. In the infinite-horizon model, the planner cannot commit to this policy if we assume a lack of commitment. Moreover, if we instead consider the social planner under commitment, this promise is time-inconsistent, as pointed out by J. Bianchi and Mendoza (2018). Therefore, this effect should be treated with caution.

The difference between the social and private shadow values of wealth, in turn, creates the wedges between the planner's optimality conditions of borrowing and investment and those of private agents. These wedges arise only when the collateral constraint binds in period 1, because the shadow values of wealth of the planner and private agents coincide when $\mu^{sp}(s) = 0$. Intuitively, when the collateral constraint binds already, the social planner tries to ameliorate the severity of the crisis by allocating resources across consumption and physical capital to push up the collateral prices, thereby relaxing the constraint.

Period 0 SP's Problem. Given that the social planner lacks the ability to commit to future policies, the planner's problem in period 0 is given by

$$\begin{aligned} \max_{\bar{c}_0, \bar{k}_1, \bar{d}_1, q_0} \quad & u(\bar{c}_0) + \mathbb{E}_0 \left[V^{sp} \left(ZA\bar{k}_1 - \bar{d}_1, \bar{k}_1 \right) \right] \\ \text{s.t.} \quad & \bar{c}_0 + \bar{k}_1 + \Psi \left(\frac{\bar{k}_1 - \bar{k}_0}{\bar{k}_0} \right) \bar{k}_0 = \bar{k}_0 + \bar{d}_1, \\ & q_0 u'(\bar{c}_0) = \mathbb{E}_0 [u'(\bar{c}_1) \{ \alpha ZA + q_1 \} + \mu^{sp} \kappa q_1], \end{aligned} \tag{26}$$

where $V^{sp}(\cdot), \bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), \bar{i}_1(s), q_1(s)$, and $\mu^{sp}(s)$, for all $s \in S$ are taken as given. The first-order conditions are given by

$$u'(\bar{c}_0) = \mathbb{E}_0 \left[V_m^{sp}(\bar{m}_1, \bar{k}_1) \right],$$

$$\left[1 + \Psi' \left(\frac{\bar{k}_1 - \bar{k}_0}{\bar{k}_0} \right) \right] u'(\bar{c}_0) = \mathbb{E}_0 \left[V_m^{sp}(\bar{m}_1, \bar{k}_1) ZA + V_k^{sp}(\bar{m}_1, \bar{k}_1) \right],$$

where $V_m^{sp}(\cdot) := \frac{\partial V^{sp}(\cdot)}{\partial \bar{m}_1}$ and $V_k^{sp}(\cdot) := \frac{\partial V^{sp}(\cdot)}{\partial \bar{k}_1}$. The envelope theorem implies that

$$V_m^{sp}(\cdot) = \lambda_1^{sp}(s) = u'(\bar{c}_1(s)) - u''(\bar{c}_1(s))q_1(s) \frac{\mu^{sp}(s)\kappa\bar{k}_1}{u'(\bar{c}_1(s))},$$

$$V_k^{sp}(\cdot) = \lambda_1^{sp}(s) \left[1 + \Psi' \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) \right] + \mu^{sp}(s)\kappa q_1(s).$$

Thus, using the envelope conditions, we obtain the social planner's Euler equation

$$u'(\bar{c}_0) = \mathbb{E}_0 \left[u'(\bar{c}_1) - u''(\bar{c}_1)q_1 \frac{\mu^{sp}\kappa\bar{k}_1}{u'(\bar{c}_1)} \right] \quad (27)$$

and the social planner's optimal investment condition

$$\left[1 + \Psi' \left(\frac{\bar{k}_1 - \bar{k}_0}{\bar{k}_0} \right) \right] u'(\bar{c}_0) =$$

$$\mathbb{E}_0 \left[u'(\bar{c}_1) \left\{ ZA + 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} + \mu^{sp}\kappa q_1 \right.$$

$$\left. - u''(\bar{c}_1)q_1 \frac{\mu^{sp}\kappa\bar{k}_1}{u'(\bar{c}_1)} \left\{ ZA + 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} \right]. \quad (28)$$

The second key difference between the social planner's equilibrium and the decentralized equilibrium follows from examining equations (27) and (17) and equations (28) and (16). First, we compare the Euler equation of the planner (27) to that of the private agents (17). In comparison with equation (17), the Euler equation of the planner (27) includes the additional term $-\mathbb{E}_0 \left[u''(\bar{c}_1(s))q_1(s) \frac{\mu^{sp}(s)\kappa\bar{k}_1}{u'(\bar{c}_1(s))} \right]$, which is positive whenever the collateral constraint is expected to bind in period 1 with positive probability. The presence of this additional term implies that the planner's expected marginal cost of borrowing in period 0, given by the right-hand side of equation (27), is higher than that of private agents. This is because the social planner internalizes that the larger debt issued in period 0 decreases the period-1 borrowing capacity in the constrained period 1–2 equilibrium where the collateral constraint is binding. This leads to the reduction of consumption in period 1 and thus exacerbates financial crises. Intuitively, because the social planner values greater period-1 consumption in the state where the collateral constraint is binding, he tries to reduce the debt in period 0. Therefore, correcting pecuniary

externalities of the entrepreneur's borrowing decisions serves to decrease the debt issued in period 0.

Importantly, this does not necessarily mean that the decentralized economy exhibits over-borrowing relative to the social planner's equilibrium in this model. As explained in Section 3.2, the interaction between externalities present in the decentralized economy matters, and whether there is over-/under-borrowing in the decentralized equilibrium depends crucially on the relative strength of two forces, one of which serves to decrease capital demand and the other of which serves to increase capital supply.

Next, we compare the optimal investment condition of the planner (28) to that of private agents (16). This is where physical capital externalities come in. Rewriting equation (28), taking into account that q_0 satisfies the asset pricing implementability constraint (18), yields the following:

$$\begin{aligned}
q_0 = & 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \\
& - \mathbb{E}_0 \left[\frac{u'(\bar{c}_1)}{u'(\bar{c}_0)} \{ZA - \alpha ZA\} \right] \\
& - \mathbb{E}_0 \left[\frac{u'(\bar{c}_1)}{u'(\bar{c}_0)} \left\{ \left[1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right] - q_1 \right\} \right] \\
& + \frac{\mathbb{E}_0 \left[u''(\bar{c}_1) q_1 \frac{\mu^{sp} \kappa \bar{k}_1}{u'(\bar{c}_1)} \left\{ 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} \right]}{u'(\bar{c}_0)} \\
& + \frac{\mathbb{E}_0 \left[u''(\bar{c}_1) q_1 \frac{\mu^{sp} \kappa \bar{k}_1}{u'(\bar{c}_1)} ZA \right]}{u'(\bar{c}_0)}
\end{aligned} \tag{29}$$

In comparison with equation (16), this expression clearly indicates that externalities present in the decentralized economy create a wedge between the optimal investment conditions of the planner and of private agents. Importantly, this expression also shows that the overall wedge can be decomposed into two parts, physical capital externalities and pecuniary externalities. The latter can be further decomposed into two parts, pecuniary externalities via capital adjustment costs and fire-sale externalities. We explain each component in what follows.

Physical Capital Externalities The term in the second line of equation (29)

$$- \mathbb{E}_0 \left[\frac{u'(\bar{c}_1)}{u'(\bar{c}_0)} \{ZA - \alpha ZA\} \right]$$

captures the distortions caused by physical capital externalities, which are discounted by the stochastic discount factor $u'(\bar{c}_1)/u'(\bar{c}_0)$. The marginal product of capital in the social planner's equilibrium is given by $Z(s)A$ and is higher than that of the decentralized equilibrium, $\alpha Z(s)A$. The difference arises because the social planner internalizes technological spillovers from capital accumulation, whereas private agents do not. Intuitively, private agents taking technological

spillovers as given undervalue the marginal product of capital, which induces them to invest less in the decentralized economy. The social planner, in contrast, internalizes technological spillovers and thus chooses larger investment than in the decentralized economy. This term serves to increase investment relative to the decentralized equilibrium because it is negative.

Pecuniary Externalities As mentioned above, the component related to pecuniary externalities can be further decomposed into two parts. Unlike physical capital externalities, the signs of the pecuniary externalities of capital investment are, in general, ambiguous.²³

(i) *Capital Adjustment Costs*: The term in the third line of equation (29)

$$-\mathbb{E}_0 \left[\frac{u'(\bar{c}_1)}{u'(\bar{c}_0)} \left\{ \left[1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right] - q_1 \right\} \right]$$

captures the distortions caused by pecuniary externalities via capital adjustment costs, which are discounted by the stochastic discount factor $u'(\bar{c}_1)/u'(\bar{c}_0)$. In this model, capital goods producers take the price of capital as given. Thus, they do not internalize the effect of their investment decisions on capital prices. Hence, with the market incompleteness, pecuniary externalities work through the price of capital not only as the collateral prices in the credit constraint of the entrepreneur, but also as output prices of capital goods producers. Because capital goods producers incur adjustment costs when producing new capital, terms associated with the capital adjustment cost appear in the above expression.

(ii) *Fire-sale Externalities*: The term in the fourth line of equation (29)

$$\frac{\mathbb{E}_0 \left[u''(\bar{c}_1) q_1 \frac{\mu^{sp} \kappa \bar{k}_1}{u'(\bar{c}_1)} \left\{ 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} \right]}{u'(\bar{c}_0)}$$

captures the distortions caused by pecuniary externalities, and is related to the so-called *fire-sale externalities*, which are normalized by the current marginal utility. This component is referred to as fire-sale externalities because it captures the possibility of fire sales in the future. To see this, notice that this term is present whenever the collateral constraint is expected to bind with positive probability in period 1. Intuitively, the social planner internalizes the effect of capital investment on the likelihood and severity of the crisis.

Lastly, the term in the fifth line of equation (29)

$$\frac{\mathbb{E}_0 \left[u''(\bar{c}_1) q_1 \frac{\mu^{sp} \kappa \bar{k}_1}{u'(\bar{c}_1)} Z A \right]}{u'(\bar{c}_0)}$$

captures the distortions caused by the combination of fire-sale externalities and physical capital externalities, in the sense that this term is present whenever there are states in which the collateral constraint is binding in period 1 and includes the marginal product of capital of the social

²³See [Dávila and Korinek \(2018\)](#) for a discussion on the difficulty of determining the sign of pecuniary externalities of capital investment.

planner rather than that of private agents. This term serves to increase investment relative to the decentralized equilibrium because it is negative, recalling that $u''(\cdot) < 0$.

Lemma 1 summarizes the above decomposition result.

Lemma 1. *The overall wedge between the optimal investment conditions of the planner and of private agents in period 0 can be decomposed into the sum of four parts, $\mathcal{G}, \mathcal{K}, \mathcal{F}, \mathcal{C}$, such that:*

$$\mathcal{G} := -\mathbb{E}_0 \left[\frac{u'(\bar{c}_1)}{u'(\bar{c}_0)} \{ZA - \alpha ZA\} \right] \quad (30)$$

$$\mathcal{K} := -\mathbb{E}_0 \left[\frac{u'(\bar{c}_1)}{u'(\bar{c}_0)} \left\{ \left[1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right] - q_1 \right\} \right] \quad (31)$$

$$\mathcal{F} := \frac{\mathbb{E}_0 \left[u''(\bar{c}_1) q_1 \frac{\mu^{sp} \bar{k}_1}{u'(\bar{c}_1)} \left\{ 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} \right]}{u'(\bar{c}_0)} \quad (32)$$

$$\mathcal{C} := \frac{\mathbb{E}_0 \left[u''(\bar{c}_1) q_1 \frac{\mu^{sp} \bar{k}_1}{u'(\bar{c}_1)} ZA \right]}{u'(\bar{c}_0)} \quad (33)$$

where $\mathcal{G}$ captures the distortions caused by growth externalities, $\mathcal{K}$ captures the distortions caused by pecuniary externalities via capital adjustment costs, $\mathcal{F}$ captures the distortions caused by fire-sale externalities, and $\mathcal{C}$ captures the distortions caused by the combination of fire-sale externalities and growth externalities.

By this decomposition, we can isolate distortions caused by physical capital externalities from those caused by pecuniary externalities. This is particularly useful in analyzing the effects of macroprudential regulation, which is intended to correct adverse financial amplification effects stemming from pecuniary externalities, not from physical capital externalities. Because the signs of the pecuniary externalities of capital investment are ambiguous, we cannot, in general, determine the sign of the overall wedge.

In summary, period 1–2 allocations in the social planner's equilibrium and in the decentralized equilibrium differ only when the collateral constraint is binding. When the constraint binds already in period 1, the planner, who internalizes the effects of his borrowing and investment decisions on the price of capital, tries to mitigate the severity of the crisis by allocating resources across consumption and physical capital in order to increase the key market price that enters the collateral constraint to relax it. The period 0 social planner's equilibrium differs from the period 0 decentralized equilibrium. In period 0, the social planner internalizes the impacts of the aggregate debt and capital investment on the probability and severity of the crisis. The decomposition of the wedge discussed earlier enables us to differentiate between the distortions caused by physical capital externalities and those caused by pecuniary externalities.

Finally, we close this section by providing a formal definition of constrained efficiency. The definition of constrained efficient equilibrium is as follows:

Definition 2. Given $(k_0, \{Z(s)\}_{s \in S})$, a constrained efficient equilibrium for this economy is defined as a vector $(\bar{c}_0, \bar{k}_0, \bar{d}_1, \bar{i}_0, q_0, \{\bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), \bar{i}_1(s), \mu^{sp}(s), q_1(s)\}_{s \in S})$ that solves the period 1–2 social planner's problem (21) and the period 0 social planner's problem (26).

3.2 Decentralization

We now show that the constrained efficient equilibrium can be decentralized with Pigouvian taxes/subsidies on debt and capital investment.²⁴²⁵ We also illustrate the normative properties of the decentralized economy. That is, we examine whether there are over-/under-borrowing and over-/under-investment in the period 0 decentralized equilibrium relative to the period 0 social planner's equilibrium. To this end, let τ_t^d and τ_t^k be the taxes charged on debt issued by the entrepreneur in period t and the production of new capital by capital goods producers in period t , respectively. We assume that the government rebates all its tax revenue to the entrepreneur in a lump-sum fashion. With the taxes in place, the budget constraints of the entrepreneur in periods 0 and 1 become

$$c_0 + q_0 k_1 = q_0 k_0 + \frac{d_1}{1 + \tau_0^d} + \Pi_0 + T_0,$$

$$c_1(s) + q_1(s) k_2(s) = m_1(s) + q_1(s) k_1 + \frac{d_2(s)}{1 + \tau_1^d(s)} + \Pi_1(s) + T_1(s), \quad \forall s$$

where $T_0 = \frac{\tau_0^d}{1 + \tau_0^d} d_1 + \tau_0^k k_1$ and $T_1(s) = \frac{\tau_1^d(s)}{1 + \tau_1^d(s)} d_2(s) + \tau_1^k(s) k_2(s)$, and the profits of capital goods producers become

$$\begin{aligned} \Pi_0 &= q_0(k_1 - k_0) - \left[\left(1 + \tau_0^k\right) k_1 - k_0 + \Psi\left(\frac{k_1 - k_0}{k_0}\right) k_0 \right], \\ \Pi_1(s) &= q_1(s)(k_2(s) - k_1) - \left[\left(1 + \tau_1^k(s)\right) k_2(s) - k_1 + \Psi\left(\frac{k_2(s) - k_1}{k_1}\right) k_1 \right]. \end{aligned}$$

The optimality conditions of private agents are now modified as follows:

$$q_0 - \tau_0^k = 1 + \Psi'\left(\frac{k_1 - k_0}{k_0}\right) \quad (34)$$

$$q_1(s) - \tau_1^k(s) = 1 + \Psi'\left(\frac{k_2(s) - k_1}{k_1}\right), \quad \forall s \quad (35)$$

$$u'(c_0) = \left(1 + \tau_0^d\right) \mathbb{E}_0[u'(c_1)] \quad (36)$$

$$u'(c_1(s)) = \left(1 + \tau_1^d(s)\right) u'(c_2(s)) + \mu(s), \quad \forall s \quad (37)$$

Comparing the social planner's optimality conditions to those of private agents in the decentralized equilibrium leads to the following proposition:

²⁴Here, we follow a Pigouvian taxation approach in dealing with externalities, which is the simplest approach taken in the literature. However, as shown by J. Bianchi (2011) and Stein (2012), there are a variety of policy instruments that would be used with equivalent results (e.g., loan-to-value ratios and capital requirements).

²⁵In what follows, we refer to these taxes/subsidies simply as taxes.

Proposition 1. *The constrained efficient equilibrium can be decentralized with Pigouvian taxes on debt and capital investment, with tax revenue rebated as a lump-sum transfer to the entrepreneur and the tax rates set such that:*

$$\tau_1^d(s) = \frac{\mu^{sp}(s)\kappa q_1(s)k_1 \left[\frac{u''(c_1(s))}{u'(c_1(s))} + \frac{u''(c_2(s))}{u'(c_2(s))} \right]}{u'(c_2(s))} \leq 0, \quad \forall s \quad (38)$$

$$\tau_1^k(s) = \frac{-\mu^{sp}(s)\kappa q_1(s)k_1 \left[\frac{u''(c_1(s))}{u'(c_1(s))} \left\{ 1 + \Psi' \left(\frac{k_2(s)-k_1}{k_1} \right) \right\} + \frac{u''(c_2(s))}{u'(c_2(s))} A \right]}{u'(c_1)} \geq 0, \quad \forall s \quad (39)$$

$$\tau_0^d = \frac{-\mathbb{E}_0 \left[u''(c_1) q_1 \frac{\mu^{sp} \kappa k_1}{u'(c_1)} \right]}{\mathbb{E}_0 [u'(c_1)]} \geq 0 \quad (40)$$

$$\tau_0^k = \mathcal{G} + \mathcal{K} + \mathcal{F} + \mathcal{C} \geq 0 \quad (41)$$

where $\mathcal{G}, \mathcal{K}, \mathcal{F}$, and $\mathcal{C}$ are defined in Lemma 1.

Proof. See Appendix B. □

Proposition 1 describes how to set Pigouvian taxes on debt and capital investment $\tau_0^d, \tau_0^k, \tau_1^d(s)$, and $\tau_1^k(s)$, for all $s \in S$ to correct pecuniary externalities and physical capital externalities that are present in the decentralized economy, as well as to implement the constrained efficient allocations. Proposition 1 also identifies the signs of most of the corrective taxes, which helps us interpret how the planner achieves constrained efficiency and explore the normative properties of the decentralized economy. Note importantly that because $\tau_0^d, \tau_0^k, \tau_1^d(s)$, and $\tau_1^k(s)$ represent the wedges between the decentralized equilibrium allocation without and with Pigouvian taxes, these optimal taxes reflect the degree of inefficiency caused by pecuniary externalities and physical capital externalities.

We first focus on Pigouvian taxes set in period 1. These corrective taxes on debt and capital investment are imposed only when the collateral constraint binds in period 1. That is, $\tau_1^d(s) = \tau_1^k(s) = 0$ when the collateral constraint does not bind. To see this, recall that $\mu^{sp} > 0$ when the collateral constraint binds, and $\mu^{sp} = 0$ when it does not. Note that a negative $\tau_1^d(s)$ means a subsidy on debt and a positive $\tau_1^k(s)$ means a tax on capital investment.

Figure 2 graphically illustrates how the social planner achieves constrained efficiency in period 1 by setting these optimal taxes. The capital supply curve, given by the optimal investment condition of capital goods producers (8), is upward-sloping, while the capital demand curve, given by the asset pricing equation of the entrepreneur (12), is downward-sloping. When the collateral constraint binds already in period 1, the planner tries to mitigate the severity of the crisis by allocating resources across consumption and physical capital in order to push up the key market price that enters the collateral constraint (i.e., capital prices), to relax it. Thus, to prop up the price of capital, the planner first sets a positive $\tau_1^k(s)$ to reduce the capital supply. This induces the capital supply curve to shift leftward, which serves to increase the price of capital. Intuitively, given capital demand, the reduction in capital supply causes the equilibrium

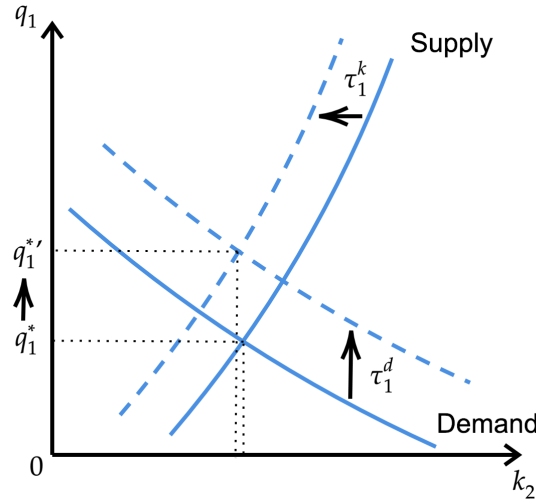


Figure 2: Pigouvian Taxes in Period 1

Notes: Illustration of how the planner mitigates the severity of the crisis when the collateral constraint binds already in period 1. The planner decreases the capital supply by setting taxes $\tau_1^k(s)$, shifting the capital supply curve leftward. He also increases the capital demand by providing subsidies $\tau_1^d(s)$, shifting the capital demand curve upward. As a result, the equilibrium price is higher than in the case without the taxes.

capital price to increase, so as to clear the capital market. Moreover, the planner subsidizes the entrepreneur by setting a negative $\tau_1^d(s)$. This, in general, increases capital demand, shifting the capital demand curve upward, leading to a further increase in the equilibrium capital price. As a result, the equilibrium price of capital is higher than in the case without the corrective taxes.

Next, we turn to Pigouvian taxes set in period 0. The optimal tax rate on debt in period 0, τ_0^d , is positive whenever the collateral constraint is expected to bind with positive probability in period 1. A Pigouvian tax τ_0^d , which corrects pecuniary externalities of the entrepreneur's borrowing decisions, serves to decrease debt issued in period 0. In contrast, the sign of the optimal tax rate on capital investment in period 0, τ_0^k , is ambiguous. However, in the presence of physical capital externalities, this tends to be negative for a large range of parameter values. Thus, a Pigouvian tax τ_0^k , that corrects physical capital externalities and pecuniary externalities of capital goods producers' investment decisions, tends to increase capital investment of capital goods producers.

Over-/Under-borrowing and Over-/Under-investment. We now explore whether over-/under-borrowing or over-/under-investment exists in the period 0 decentralized equilibrium relative to the period 0 social planner's equilibrium. We show that these depend crucially on the relative strength of the following two forces: (i) the force to reduce capital demand, induced by pecuniary externalities of the entrepreneur's borrowing decisions, and (ii) the force that tends to increase capital supply, induced by physical capital externalities and pecuniary externalities of capital goods producers' investment decisions. The externality that generates

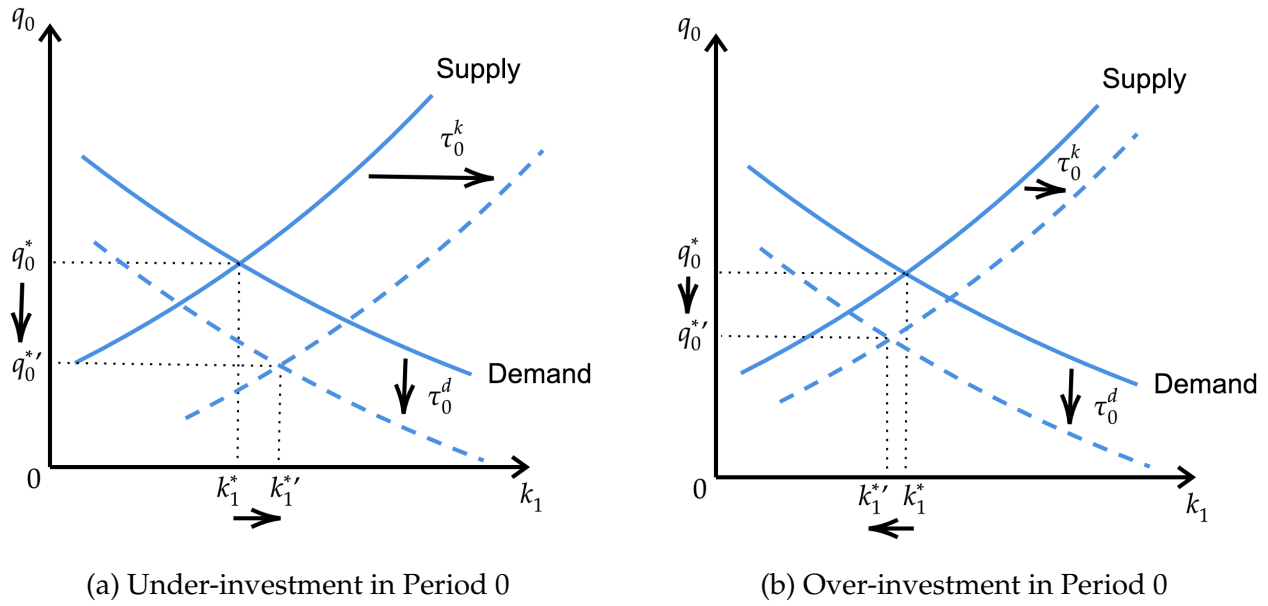


Figure 3: Pigouvian Taxes in Period 0

Notes: Illustration of the interaction between externalities corrected by τ_0^d and τ_0^k . The dashed lines show how the planner increases the capital supply and reduces the capital demand by setting Pigouvian taxes τ_0^k and τ_0^d , respectively. The intersection of the bold lines represents the period 0 decentralized equilibrium, while the intersection of the dashed lines represents the period 0 social planner's equilibrium. Depending on the relative strength of these two forces, the decentralized economy exhibits over-/under-investment in period 0.

the former force is corrected by τ_0^d , while those generating the latter force are corrected by τ_0^k .

To better understand the interaction between externalities in period 0, we illustrate possible cases in our model economy. Figure 3 graphically illustrates the interaction between the externalities corrected by the Pigouvian taxes τ_0^d and τ_0^k . The capital supply curve, given by the optimal investment condition of capital goods producers (16), is upward-sloping. We draw the downward-sloping capital demand curve, given by the asset pricing equation of the entrepreneur (18), by assuming that the tightness of the collateral constraint κ is sufficiently small.²⁶

Figure 3a illustrates the case in which the increase in the capital supply caused by a negative τ_0^k is large enough to yield under-investment in the decentralized economy. In this case, the positive externalities of capital investment are so strong that correcting these externalities by a negative τ_0^k increases capital supply significantly, leading to a large rightward shift of the capital supply curve. Thus even though capital demand is suppressed by a tax on debt τ_0^d , a sharp decline in the equilibrium capital price allows the entrepreneur to purchase more capital than

²⁶We set the baseline parameters so that this is true in our model economy. Moreover, as implied by Mendoza (2010) and J. Bianchi (2011), the parameter κ representing the tightness of the credit constraint should be set to be sufficiently small so as to match the frequency of financial crises in the emerging economies.

in the case without the taxes. As a result, the decentralized economy exhibits under-investment relative to the constrained efficient equilibrium. Furthermore, in this case, the decentralized economy tends to exhibit under-borrowing. This is because at very low price levels, the entrepreneur may find it more beneficial to purchase newly created capital by issuing more debt even though he faces higher effective costs of borrowing because of a positive τ_0^d .

Figure 3b illustrates the case in which the force to suppress the capital demand by a positive τ_0^d outweighs the force to increase the capital supply caused by a negative τ_0^k , so that the decentralized economy exhibits over-investment in period 0. In this case, there tends to be over-borrowing in the decentralized economy. This is because the benefit to issue more debt in period 0 described above is not so large that the effect of τ_0^d which serves to decrease debt in period 0 dominates. Intuitively, the precautionary motive of the social planner to avoid severe crises exceeds the effect caused by correcting externalities of capital investment.

3.3 Numerical Illustration

This section presents the quantitative implications of the model by conducting simple numerical simulations. We first focus on the financial crisis dynamics and how the allocations are different between the decentralized equilibrium and the social planner's equilibrium for the baseline parameter setting. We then explore the effectiveness of Pigouvian taxes in reducing the likelihood and severity of financial crises. As explained earlier, there are pecuniary externalities and physical capital externalities in the decentralized economy, and the social planner corrects both externalities by setting taxes on debt and capital investment. Practically, however, macroprudential regulation aims at containing systemic risk in the financial sector and is intended to deal with adverse financial amplification effects of financial crises that are related to pecuniary externalities, not to physical capital externalities. Thus, in addition to the social planner's equilibrium, we also consider the planner's equilibrium where the planner corrects only pecuniary externalities.²⁷

In the same way as in Section 2.3, we conduct numerical simulations of the following three cases: (i) the decentralized equilibrium (DE), (ii) the social planner's equilibrium (SP equilibrium), and (iii) the planner's equilibrium in which the planner corrects only pecuniary externalities (SPPE equilibrium).²⁸ Focusing on the case where the equilibrium is constrained in periods 1 and 2, the resulting allocations are summarized in Table 5.

Crisis Dynamics. We first compare the SP equilibrium to the DE. The first important result

²⁷In this case, we let the marginal product of capital of the social planner remain $\alpha Z(s)A$, which is the same as the marginal product of capital of private agents. That is, the planner does not correct growth externalities. This implies that the term $\mathcal{G}$ in equation (41) vanishes. There is also a slight modification in the term $\mathcal{C}$ in equation (41), or equation (33), i.e., ZA in the bracket becomes αZA .

²⁸We use the same functional form for preferences and parameter values as presented in Table 1, and assume also that the collateral constraint binds when Z^l is realized and not when Z^h is realized.

Table 5: Comparison of Crisis Dynamics

	Decentralized equilibrium (DE)	Social planner's equilibrium (SP eq)	SP correcting only pecuniary externalities
d_1	4.218	5.069	4.027
d_2^c	0.193	0.153	0.245
$n_0(=k_1 - k_0)$	0.148	0.882	0.089
$n_1^c(=k_2^c - k_1)$	-0.485	-0.581	-0.506
c_0	4.060	3.852	3.934
c_1^c	1.699	1.626	1.878
c_2^c	6.802	7.798	6.629
q_0	1.138	0.870	1.047
q_1^c	0.094	0.065	0.120

Notes: The DE corresponds to the one explained in Section 2.2. The SP equilibrium corresponds to the allocation in which the social planner corrects both pecuniary externalities and growth externalities. The SP correcting only pecuniary externalities corresponds to the allocation in which the planner corrects only pecuniary externalities.

of this quantitative analysis is that the social planner borrows more and invests more than private agents in period 0, implying that there are under-borrowing and under-investment in the DE (compare the first line and the third line of Table 5). This result contrasts with most of the existing literature emphasizing over-borrowing in the DE and is in line with the findings of Benigno et al. (2013) and Schmitt-Grohé and Uribe (2018). However, in contrast to these studies, the decentralized economy in our model also exhibits under-investment, in addition to under-borrowing.

Next, we focus on the SPPE equilibrium. In this case, the planner issues less debt and invests less in period 0 than in the DE. Moreover, the crisis dynamics is less severe than in the DE, in the sense that declines in investment and drops in capital prices are moderate and the degree of failure of consumption smoothing is also mild. Hence, the planner in this case succeeds in mitigating the severity of financial crises by reducing debt and investment in period 0.

In summary, this simple numerical simulation provides two key results. First, there are under-borrowing and under-investment in the DE in period 0 relative to the SP equilibrium, while the planner correcting only pecuniary externalities borrows less and invests less in period 0 than in the DE. Second, the crisis dynamics is less severe in the SPPE equilibrium than in the DE.

Table 6 illustrates additional statistics that summarizes the effectiveness of the taxes on debt and capital investment. Table 6 shows that compared to the DE, the threshold value of $Z(s)$ is higher in the SP equilibrium, while that is lower in the SPPE equilibrium. Here, the threshold

Table 6: Summary Statistics

	Decentralized equilibrium (DE)	Social planner's equilibrium (SP eq)	SP correcting only pecuniary externalities
Threshold value of $Z(s)$	0.961	0.989	0.943
Welfare gains		3.1%	-0.4%
Tax on d_1 (τ_0^d)		15.1%	18.8%
Tax on d_2 (τ_1^d)		-650%	-577%
Tax on k_1 (τ_0^k)		-86.8%	-3.8%
<i>growth externalities</i> ($\mathcal{G}$)		-71.9%	0%
<i>capital adjustment</i> ($\mathcal{K}$)		-0.4%	2.4%
<i>fire-sale externalities</i> ($\mathcal{F}$)		-0.7%	-0.6%
<i>combination</i> ($\mathcal{C}$)		-13.8%	-5.5%
Tax on k_2 (τ_1^k)		7.2%	14.5%

Notes: The threshold value of $Z(s)$ is defined as the value of $Z(s)$ that equates the LHS and the RHS of the threshold condition (15). Welfare gains are measured as compensating consumption variations that equate the expected lifetime utility for the DE and SP equilibrium/SPPE equilibrium. The taxes on debt and capital investment are calculated from equations (38) – (41) in Proposition 1. In a tax on k_1 , ‘growth externalities’ corresponds to the term $\mathcal{G}$ in equation (41), ‘capital adjustment’ corresponds to the term $\mathcal{K}$, ‘fire-sale externalities’ corresponds to the term $\mathcal{F}$, and ‘combination’ corresponds to the term $\mathcal{C}$.

value of $Z(s)$ is defined as the value of $Z(s)$ that equates the left-hand side and the right-hand side of the threshold condition (15) and is considered as measuring the likelihood of financial crises.²⁹ The low threshold value of $Z(s)$ means less likelihood of crises because the entrepreneur is less likely to face the binding collateral constraint even when the low value of $Z(s)$ is realized. Therefore, relative to the DE, the probability of financial crises is higher in the SP equilibrium and lower in the SPPE equilibrium.

Welfare Effects. Next, we explore the welfare implications of the taxes on debt and capital investment. The welfare gains from correcting externalities are calculated as compensating consumption variations for each state of nature that make entrepreneurs indifferent between the allocations of the decentralized equilibrium and those attained by the social planner. Formally, for a given initial state k_0 , the welfare gains of Pigouvian taxes on debt and capital investment are computed as the value of $\gamma(k_0)$ that satisfies the condition:

$$\mathbb{E}_0 \sum_{t=0}^2 \left[u \left(c_t^{DE} (1 + \gamma) \right) \right] = \mathbb{E}_0 \sum_{t=0}^2 \left[u \left(c_t^{SP} \right) \right]$$

²⁹This is because the likelihood that the collateral constraint binds varies depending on the threshold value of $Z(s)$.

where the superscript DE denotes allocations in the DE and the superscript SP denotes the social planner's allocations. Here, the social planner's allocations can be either the SP equilibrium or the SPPE equilibrium.

Importantly, Table 6 shows that the SP equilibrium achieves the welfare gains of 3.1% of permanent consumption, while the SPPE equilibrium attains the negative welfare gains of -0.4%. This result, together with the results described in Table 5, provides one of the most important results of this paper: Although macroprudential regulation that corrects only pecuniary externalities is effective in reducing the likelihood of a financial crisis and alleviating its severity, it is welfare-reducing because it reduces productive investment, thereby impeding economic growth. Put differently, the growth-reducing effect of the regulation exceeds the growth-enhancing effect, and the cost of hindering growth outweighs the benefit of securing financial stability, leading to a welfare-reducing result.

Tax Rates. Lastly, we illustrate tax rates on debt and capital investment. We first present the optimal tax rates in the SP equilibrium and then the tax rates in the SPPE equilibrium.³⁰ The optimal tax rates on debt and capital investment in period 0, τ_0^d and τ_0^k , are 15.1% and -86.8%, respectively. A large subsidy on capital investment is mainly attributed to the components dealing with growth externalities, as shown in Table 6.³¹ This large subsidy on capital investment and a relatively low tax rate on debt allow us to interpret the period-0 SP equilibrium summarized in Table 5 as follows. A huge subsidy on capital investment significantly increases capital supply, leading to a large rightward shift of the capital supply curve. A tax on debt decreases capital demand, shifting the capital demand curve downward. Since the increase in capital supply is relatively larger than the decline in capital demand, investment in the SP equilibrium is larger than in the DE. Moreover, drops in the equilibrium capital price are large enough that the entrepreneur finds it beneficial to purchase newly created capital more by issuing more debt even though he faces higher effective costs of borrowing because of a positive τ_0^d . As a result, there are under-investment and under-borrowing in the DE relative to the SP equilibrium.

The optimal taxes when the crises occur are given by $\tau_1^d = -650\%$ and $\tau_1^k = 14.5\%$. Here, the optimal subsidy on debt when the crisis occurs takes unrealistically high values, and one might interpret this result as follows. Financial crises are so severe in our model economy that the government takes the emergency fiscal packages of subsidizing the entrepreneur to mitigate a vicious cycle of a financial crisis.³² The intuition for the constrained period 1-2 social planner's equilibrium summarized in Table 5 is as follows. The planner imposes a tax on capital investment of capital goods producers to reduce capital supply, leading to the leftward shift

³⁰Note that taxes set to correct only pecuniary externalities are not optimal.

³¹Recall that the component dealing with both fire-sale externalities and physical capital externalities, given by the term $\mathcal{C}$ in equation (41), partially includes the effect of dealing with growth externalities. It accounts for -13.8% out of -86.8%.

³²In the present simple three-period model, it is hard to generate realistic values of the optimal subsidy on debt for plausible parameter values, and thus τ_1^d takes unrealistically high values.

of the capital supply curve. The planner also subsidizes the entrepreneur to increase capital demand, shifting the capital demand curve upward. As a result, the equilibrium capital price rises and this relaxes the collateral constraint, mitigating the severity of the crisis.

We now focus on the tax rates of the SPPE equilibrium. The important difference from the optimal tax rates is that a subsidy on capital investment in period 0 is much lower than in the SP equilibrium. The tax rate on capital investment in period 0 of the planner correcting only pecuniary externalities is given by $\tau_0^k = -3.8\%$, while that of the social planner correcting both externalities is $\tau_0^k = -86.8\%$. Given that a tax rate on debt is set similarly, this relatively low subsidy on capital investment helps us interpret the SPPE equilibrium as follows. An increase in the capital supply by a subsidy on capital investment is relatively smaller than the decrease in the capital demand by a tax on debt. As a result, investment in the SPPE equilibrium is lower than in the DE. Furthermore, the decline in the equilibrium capital price is not so large that the entrepreneur does not find it beneficial to invest more by borrowing more. Thus, the entrepreneur facing higher effective costs of borrowing due to a positive τ_0^d issues less debt than in the DE. Other tax rates are almost similar to the optimal ones, and thus we omit the explanation.

4 Conclusion

This paper analyzes the impact of macroprudential regulation on economic growth and its welfare implications. To this end, we introduce technological spillovers and asymmetric capital adjustment costs into a model of financial crises. A price-dependent collateral constraint and technological spillovers introduce pecuniary externalities and growth externalities into the model. We show that a wedge between the privately and socially optimal allocations induced by these two externalities can be decomposed into each part. This decomposition allows us to obtain the following results.

Our main result is that macroprudential regulation that corrects only pecuniary externalities is welfare-reducing for a large range of parameter values. The growth-reducing effect of macroprudential regulation exceeds the growth-enhancing effect, thereby hindering economic growth. The regulation, on the other hand, reduces the likelihood of financial crises and ameliorates the severity of the crises. Therefore, there is a trade-off between economic growth and financial stability. In our model, the cost of hindering growth outweighs the benefit of financial stability, leading to a welfare-reducing result. Moreover, we show that whether the decentralized economy exhibits over-/under-borrowing or over-/under-investment depends on the relative strength of the following two forces: (i) the force to reduce capital demand in the planner's allocation, induced by pecuniary externalities of the private agent's borrowing decisions, and (ii) the force that tends to increase capital supply, induced by physical capital externalities and pecuniary externalities of private agents' investment decisions. We find that the decentralized economy exhibits under-borrowing and under-investment for a large range of parameter

values.

Importantly, since this result is derived from the simple three-period model, the conclusion should be discounted to some extent. The main virtue of the analysis in this paper is that the possible mechanisms we expected are indeed present in the model, and hence the infinite-horizon model that exhibits the flavor of this three-period model is likely to be an appropriate model for answering the research questions.

Naturally, the next step is to construct the infinite-horizon, full-blown quantitative model that generalizes the above three-period model. The main difficulty associated with it is to come up with a plausible numerical algorithm and solve the model numerically. In this regard, it might be promising to consider the diminishing returns to investment due to [Barlevy \(2004\)](#), instead of partial irreversibility of investment considered in this paper. The former is another possible way to connect volatility with economic growth and is much simpler than the latter. This might allow us to numerically solve the infinite-horizon model more easily.

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Appendix

A Equilibrium Conditions

In this section, we fully describe the equilibrium conditions of the model. We first focus on the decentralized equilibrium discussed in Section 2.2, and then on the social planner's equilibrium discussed in Section 3.1.

A.1 Decentralized Equilibrium Conditions

As in the text, we solve for the decentralized equilibrium via backward induction. The period 1–2 equilibrium conditions are:

$$q_1(s) = 1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) \quad (\text{A.1})$$

$$u'(c_1(s)) = u'(c_2(s)) + \mu(s) \quad (\text{A.2})$$

$$q_1(s) = \frac{u'(c_2(s))}{u'(c_1(s))} A \quad (\text{A.3})$$

$$c_1(s) + k_2(s) + \Psi \left(\frac{k_2(s) - k_1}{k_1} \right) k_1 = \underbrace{m_1(s)}_{=Z(s)Ak_1 - d_1} + k_1 + d_2(s) \quad (\text{A.4})$$

$$c_2(s) + d_2(s) = Ak_2(s) \quad (\text{A.5})$$

$$d_2(s) \leq \kappa q_1(s) k_1 \quad (\text{A.6})$$

$$\mu(s) \geq 0 \quad (\text{A.7})$$

$$\mu(s)(\kappa q_1(s) k_1 - d_2(s)) = 0 \quad (\text{A.8})$$

where equation (A.1) is the optimal investment condition of capital goods producers, (A.2) the Euler equation of the entrepreneur, (A.3) the asset pricing equation of the entrepreneur, (A.4) the resource constraint in period 1, (A.5) the resource constraint in period 2, (A.6) the collateral constraint, and (A.7) and (A.8) the complementary slackness conditions. Note that in period 1–2 equilibrium conditions, m_1 and k_1 are regarded as state variables (the state of nature s is realized at the beginning of period 1). Equations (A.1) – (A.8) jointly determine six endogenous variables $c_1(s), c_2(s), k_2(s), d_2(s), q_1(s), \mu(s)$. Once these variables are determined, $i_1(s)$ is determined according to $i_1(s) = k_2(s) - k_1 + \Psi \left(\frac{k_2(s) - k_1}{k_1} \right) k_1$. Depending on whether the collateral constraint binds, the period 1–2 equilibrium becomes the period 1–2 unconstrained equilibrium or the period 1–2 constrained equilibrium.

Unconstrained Period 1–2 Equilibrium

In this case, it is the case that $d_2(s) < \kappa q_1(s) k_1$ so that $\mu(s) = 0$ by the complementary slackness condition (A.8). This implies that equation (A.2) becomes:

$$u'(c_1(s)) = u'(c_2(s)) \quad (\text{A.9})$$

This then implies that

$$c_1(s) = c_2(s). \quad (\text{A.10})$$

Then equation (A.9) implies that equation (A.3) becomes

$$q_1(s) = A. \quad (\text{A.11})$$

Equation (A.11) determines $q_1(s)$. Combining equations (A.11) and (A.1), we obtain:

$$1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) = A \quad (\text{A.12})$$

Thus, $k_2(s)$ is determined so that equation (A.12) holds. Combining equations (A.10), (A.4), and (A.5) yields

$$c_1(s) = c_2(s) = \frac{1}{2} \left[m_1(s) + k_1 + Ak_2(s) - k_2(s) - \Psi \left(\frac{k_2(s) - k_1}{k_1} \right) k_1 \right]. \quad (\text{A.13})$$

Lastly, combining equations (A.5) and (A.13), we obtain:

$$d_2(s) = Ak_2(s) - \frac{1}{2} \left[m_1(s) + k_1 + Ak_2(s) - k_2(s) - \Psi \left(\frac{k_2(s) - k_1}{k_1} \right) k_1 \right] \quad (\text{A.14})$$

In summary, in the unconstrained period 1–2 equilibrium, we have $\mu(s) = 0$, and equation (A.12) determines $k_2(s)$, equation (A.13) determines $c_1(s)$ and $c_2(s)$, equation (A.14) determines $d_2(s)$, and equation (A.11) determines $q_1(s)$.

Constrained Period 1–2 Equilibrium

In this case, equation (A.6) becomes $d_2(s) = \kappa q_1(s) k_1$. Substituting equation (A.1) into this equation gives

$$d_2(s) = \kappa \left[1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) \right] k_1 \quad (\text{A.15})$$

Substituting equation (A.15) into equations (A.4) and (A.5), we obtain:

$$c_1(s) = m_1(s) + k_1 + \kappa \left[1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) \right] k_1 - k_2(s) - \Psi \left(\frac{k_2(s) - k_1}{k_1} \right) k_1 \quad (\text{A.16})$$

$$c_2(s) = Ak_2(s) - \kappa \left[1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) \right] k_1 \quad (\text{A.17})$$

We also have from (A.2) and (A.7),

$$\mu(s) = u'(c_1(s)) - u'(c_2(s)) \geq 0. \quad (\text{A.18})$$

Hence we have expressed endogenous variables $c_1(s), c_2(s), d_2(s), q_1(s), \mu(s)$ as a function of $k_2(s)$. Lastly, substituting equation (A.1) into equation (A.3), we get:

$$1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) = \frac{u'(c_2(s))}{u'(c_1(s))} A \quad (\text{A.19})$$

where $c_1(s)$ and $c_2(s)$ are given by equations (A.16) and (A.17), respectively. Therefore, once $k_2(s)$ is determined so that equation (A.19) holds, all the other endogenous variables are also determined according to equations (A.15) – (A.18).

Next, we describe the period 0 equilibrium conditions. The period 0 equilibrium conditions are:

$$q_0 = 1 + \Psi' \left(\frac{k_1 - k_0}{k_0} \right) \quad (\text{A.20})$$

$$u'(c_0) = \mathbb{E}_0 [u'(c_1)] \quad (\text{A.21})$$

$$q_0 u'(c_0) = \mathbb{E}_0 \left[u'(c_1) \left\{ \underbrace{\alpha Z A_1 k_1^{\alpha-1}}_{=\alpha Z A} + q_1 \right\} + \mu \kappa q_1 \right] \quad (\text{A.22})$$

$$c_0 + k_1 + \Psi \left(\frac{k_1 - k_0}{k_0} \right) k_0 = k_0 + d_1 \quad (\text{A.23})$$

where equation (A.20) is the optimal investment condition of capital goods producers, (A.21) the Euler equation of the entrepreneur, (A.22) the asset pricing equation of the entrepreneur, (A.23) the resource constraint in period 0. Given variables determined in the period 1–2 equilibrium, equations (A.20) – (A.23) jointly determine four endogenous variables c_0, k_1, d_1, q_0 . Once these variables are determined, i_0 is determined according to $i_0 = k_1 - k_0 + \Psi \left(\frac{k_1 - k_0}{k_0} \right) k_0$.

In summary, given the initial capital stock k_0 and the stochastic process $\{Z(s)\}_{s \in S}$, equations (A.1) – (A.8) and (A.20) – (A.23) determine $(c_0, k_1, d_1, i_0, q_0, \{c_1(s), c_2(s), k_2(s), d_2(s), i_1(s), \mu(s), q_1(s)\}_{s \in S})$.

A.2 Social Planner's Equilibrium Conditions

As in the text, we solve for the social planner's equilibrium via backward induction. The period 1–2 equilibrium conditions are:

$$u'(\bar{c}_1(s)) = u'(\bar{c}_2(s)) + \mu^{sp}(s) + \mu^{sp}(s)\kappa q_1(s)\bar{k}_1 \left[\frac{u''(\bar{c}_1(s))}{u'(\bar{c}_1(s))} + \frac{u''(\bar{c}_2(s))}{u'(\bar{c}_2(s))} \right] \quad (\text{A.24})$$

$$1 + \Psi' \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) = \frac{u'(\bar{c}_2(s))}{u'(\bar{c}_1(s))} A + \frac{\mu^{sp}(s)\kappa q_1(s)\bar{k}_1 \left[\frac{u''(\bar{c}_1(s))}{u'(\bar{c}_1(s))} \left\{ 1 + \Psi' \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) \right\} + \frac{u''(\bar{c}_2(s))}{u'(\bar{c}_2(s))} A \right]}{u'(\bar{c}_1(s))} \quad (\text{A.25})$$

$$q_1(s) = \frac{u'(\bar{c}_2(s))}{u'(\bar{c}_1(s))} A \quad (\text{A.26})$$

$$\bar{c}_1(s) = \underbrace{m_1(s)}_{Z(s)A\bar{k}_1 - \bar{d}_1} + \bar{k}_1 + \bar{d}_2(s) - \left[\bar{k}_2(s) + \Psi \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) \bar{k}_1 \right] \quad (\text{A.27})$$

$$\bar{c}_2(s) + \bar{d}_2(s) = A\bar{k}_2(s) \quad (\text{A.28})$$

$$\bar{d}_2(s) \leq \kappa q_1(s)\bar{k}_1 \quad (\text{A.29})$$

$$\mu^{sp}(s) \geq 0 \quad (\text{A.30})$$

$$\mu^{sp}(s)(\kappa q_1(s)\bar{k}_1 - \bar{d}_2(s)) = 0 \quad (\text{A.31})$$

where equation (A.24) is the planner's Euler equation, (A.25) the optimal investment condition of the planner, (A.26) the asset pricing implementability constraint, (A.27) the resource constraint in period 1, (A.28) the resource constraint in period 2, (A.29) the collateral constraint, and (A.30) and (A.31) the complementary slackness condition. Equations (A.24) – (A.31) jointly determine six endogenous variables $\bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), q_1(s), \mu^{sp}(s)$. Once these variables are determined, $i_1(s)$ is determined according to $i_1(s) = \bar{k}_2(s) - \bar{k}_1 + \Psi \left(\frac{\bar{k}_2(s) - \bar{k}_1}{\bar{k}_1} \right) \bar{k}_1$. As in the decentralized equilibrium, the period 1–2 equilibrium becomes the period 1–2 unconstrained equilibrium or the period 1–2 constrained equilibrium, depending on whether the collateral constraint binds or not.

Unconstrained Period 1–2 Equilibrium

Same as in the decentralized case.

Constrained Period 1–2 Equilibrium

In this case, (A.29) becomes:

$$\bar{d}_2(s) = \kappa q_1(s)\bar{k}_1 \quad (\text{A.32})$$

Then equations (A.24) – (A.28), (A.30), and (A.32) jointly determine $\bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), q_1(s)$, and $\mu^{sp}(s)$.

Next, we describe the period 0 equilibrium conditions. As in the text, in period 0 the social planner takes as given $V^{sp}(\cdot), \bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), q_1(s)$, and $\mu^{sp}(s)$, for all $s \in S$.

The period 0 equilibrium conditions are:

$$u'(\bar{c}_0) = \mathbb{E}_0 \left[u'(\bar{c}_1) - u''(\bar{c}_1) q_1 \frac{\mu^{sp} \kappa \bar{k}_1}{u'(\bar{c}_1)} \right] \quad (\text{A.33})$$

$$\begin{aligned} \left[1 + \Psi' \left(\frac{\bar{k}_1 - \bar{k}_0}{\bar{k}_0} \right) \right] u'(\bar{c}_0) = \\ \mathbb{E}_0 \left[u'(\bar{c}_1) \left\{ ZA + 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} + \mu^{sp} \kappa q_1 \right. \\ \left. - u''(\bar{c}_1) q_1 \frac{\mu^{sp} \kappa \bar{k}_1}{u'(\bar{c}_1)} \left\{ ZA + 1 + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) + \Psi' \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) - \Psi \left(\frac{\bar{k}_2 - \bar{k}_1}{\bar{k}_1} \right) \right\} \right] \end{aligned} \quad (\text{A.34})$$

$$q_0 u'(\bar{c}_0) = \mathbb{E}_0 [u'(\bar{c}_1) \{ \alpha ZA + q_1 \} + \mu^{sp} \kappa q_1] \quad (\text{A.35})$$

$$\bar{c}_0 + \bar{k}_1 + \Psi \left(\frac{\bar{k}_1 - \bar{k}_0}{\bar{k}_0} \right) \bar{k}_0 = \bar{k}_0 + \bar{d}_1 \quad (\text{A.36})$$

where equation (A.33) is the planner's Euler equation, (A.34) the optimal investment condition of the planner, (A.35) asset pricing implementability constraint, (A.36) the resource constraint in period 0. Given variables determined in the period 1–2 equilibrium, equations (A.33) – (A.36) jointly determine four endogenous variables $\bar{c}_0, \bar{k}_1, \bar{d}_1, q_0$. Once these variables are determined, $\bar{i}_0$ is determined according to $\bar{i}_0 = \bar{k}_1 - \bar{k}_0 + \Psi \left(\frac{\bar{k}_1 - \bar{k}_0}{\bar{k}_0} \right) \bar{k}_0$.

In summary, given the initial capital stock $\bar{k}_0$ and the stochastic process $\{Z(s)\}_{s \in S}$, equations (A.24) – (A.31) and (A.33) – (A.36) determine $(\bar{c}_0, \bar{k}_1, \bar{d}_1, \bar{i}_0, q_0, \{\bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), \bar{i}_1(s), \mu^{sp}(s), q_1(s)\}_{s \in S})$.

B Proof of Proposition 1

Define taxes as follows:

$$1 + \tau_1^d(s) = \frac{u'(c_2(s)) + \mu^{sp}(s)\kappa q_1(s)k_1 \left[\frac{u''(c_1(s))}{u'(c_1(s))} + \frac{u''(c_2(s))}{u'(c_2(s))} \right]}{u'(c_2(s))}, \quad \forall s \quad (\text{B.1})$$

$$\tau_1^k(s) = \frac{-\mu^{sp}(s)\kappa q_1(s)k_1 \left[\frac{u''(c_1(s))}{u'(c_1(s))} \left\{ 1 + \Psi' \left(\frac{k_2(s) - k_1}{k_1} \right) \right\} + \frac{u''(c_2(s))}{u'(c_2(s))} A \right]}{u'(c_1)}, \quad \forall s \quad (\text{B.2})$$

$$1 + \tau_0^d = \frac{\mathbb{E}_0[u'(c_1)] - \mathbb{E}_0 \left[u''(c_1) q_1 \frac{\mu^{sp} \kappa k_1}{u'(c_1)} \right]}{\mathbb{E}_0[u'(c_1)]} \quad (\text{B.3})$$

$$\begin{aligned} \tau_0^k = & -\mathbb{E}_0 \left[\frac{u'(c_1)}{u'(c_0)} \{ZA - \alpha ZA\} \right] \\ & - \mathbb{E}_0 \left[\frac{u'(c_1)}{u'(c_0)} \left\{ \left[1 + \Psi' \left(\frac{k_2 - k_1}{k_1} \right) + \Psi' \left(\frac{k_2 - k_1}{k_1} \right) \left(\frac{k_2 - k_1}{k_1} \right) - \Psi \left(\frac{k_2 - k_1}{k_1} \right) \right] - q_1 \right\} \right] \\ & + \frac{\mathbb{E}_0 \left[u''(c_1) q_1 \frac{\mu^{sp} \kappa k_1}{u'(c_1)} \left\{ 1 + \Psi' \left(\frac{k_2 - k_1}{k_1} \right) + \Psi' \left(\frac{k_2 - k_1}{k_1} \right) \left(\frac{k_2 - k_1}{k_1} \right) - \Psi \left(\frac{k_2 - k_1}{k_1} \right) \right\} \right]}{u'(c_0)} \\ & + \frac{\mathbb{E}_0 \left[u''(c_1) q_1 \frac{\mu^{sp} \kappa k_1}{u'(c_1)} ZA \right]}{u'(c_0)} \end{aligned} \quad (\text{B.4})$$

The constrained efficient equilibrium is characterized by a vector

$\left(\bar{c}_0, \bar{k}_0, \bar{d}_1, \bar{i}_0, q_0, \left\{ \bar{c}_1(s), \bar{c}_2(s), \bar{k}_2(s), \bar{d}_2(s), \bar{i}_1(s), \mu^{sp}(s), q_1(s) \right\}_{s \in S} \right)$ that satisfies a set of conditions (A.24) – (A.31) and (A.33) – (A.36).

The decentralized equilibrium with taxes on debt and capital investment is characterized by a vector $(c_0, k_0, d_1, i_0, q_0, \{c_1(s), c_2(s), k_2(s), d_2(s), i_1(s), \mu(s), q_1(s)\}_{s \in S})$ that satisfies (35), (37), (A.3) – (A.8), (34), (36), (A.22), and (A.23). Using the expressions for the taxes (B.1) – (B.4) yields conditions (A.24), (A.25), (A.33), and (29). Note that the combination of (A.22) and (29) are equivalent to that of (A.34) and (A.35). Therefore, the conditions characterizing the decentralized equilibrium with taxes on debt and capital investment are identical to those characterizing the constrained efficient equilibrium.

C Computational Algorithm

This section provides the computational algorithm used in numerical simulation in Sections 2.3 and 3.3. In what follows, we present the descriptions by focusing on the decentralized equilibrium. Notice that variables in the period 1–2 equilibrium are determined once values of the state variables $m_1(s)$ and k_1 are determined. Thus, solving the simultaneous equations (19) and (20) for k_1 and d_1 essentially means solving for the decentralized equilibrium. In the simultaneous equations (19) and (20), variables in the period 1–2 equilibrium can all be expressed in terms of k_2^u and k_2^c , and hence the simultaneous equations (19) and (20) include four endogenous variables (d_1, k_1, k_2^u, k_2^c) . However, k_2^u and k_2^c cannot be explicitly solved in terms of the

state variables in period 1 (see equations (A.12) and (A.19)). Thus complication arises when solving the simultaneous equations (19) and (20) for k_1 and d_1 .

To deal with this issue, we use the following computational algorithm to solve for a competitive equilibrium:

1. Set the tolerance parameter ε^k and ε^d .
2. Guess the values of k_1^{guess} and d_1^{guess} .
3. Compute k_2^u and k_2^c according to equations (A.12) and (A.19), respectively, using k_1^{guess} and d_1^{guess} .
4. Solve the simultaneous equations (19) and (20) for k_1 and d_1 , using k_2^u and k_2^c computed in Step 3.
5. Evaluate convergence. If $|k_1^{guess} - k_1| < \varepsilon^k$ and $|d_1^{guess} - d_1| < \varepsilon^d$, then we have found a competitive equilibrium. Otherwise, update the guess, $k_{1new}^{guess} = \lambda k_1 + (1 - \lambda)k_1^{guess}$ and $d_{1new}^{guess} = \lambda d_1 + (1 - \lambda)d_1^{guess}$ with $\lambda \in (0, 1]$, and go back to Step 3.